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Harmon et al.

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(54) **VEHICLE CONSOLE HAVING COOLER
ARRANGEMENT**

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patent is extended or adjusted under 35
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B60N 3/10 (2006.01)
B60N 3/00 (2006.01)

(52) **U.S. Cl.**
CPC **B60N 3/103** (2013.01); **B60N 3/002**
(2013.01); **B60N 3/104** (2013.01)

(58) **Field of Classification Search**
CPC B60N 3/103; B60N 3/104; B60N 3/002
USPC 224/400
See application file for complete search history.

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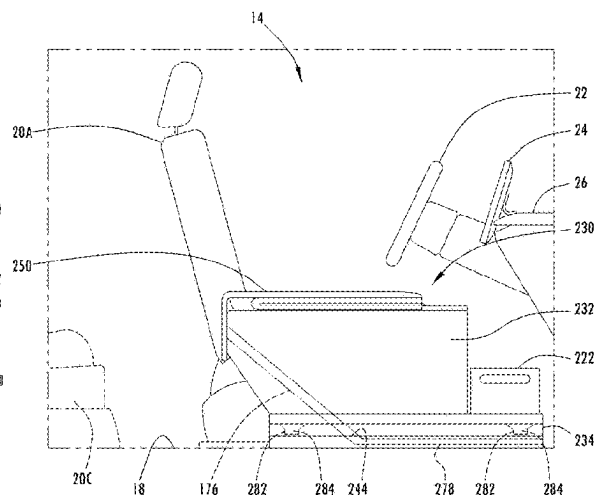
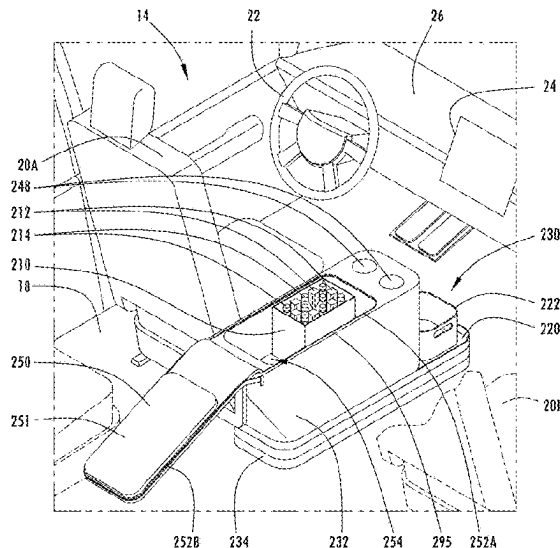
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Heneveld LLP

(57) **ABSTRACT**

A vehicle console including a base, a housing supported on
the base and comprising a storage compartment, the storage
compartment having side walls in an access opening, and a
lid operatively coupled to the access opening of the housing
and comprising a zipper for connecting the lid to the top end
of the module, wherein the zipper may be unzipped to
expose the access opening and may be zipped to close the lid
on the access opening.

16 Claims, 36 Drawing Sheets



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U.S. Appl. No. 18/085,832 (4 pages of filing receipt, 30 pages of application and 36 pages of drawings), filed Dec. 21, 2022, entitled "Vehicle Media Console."

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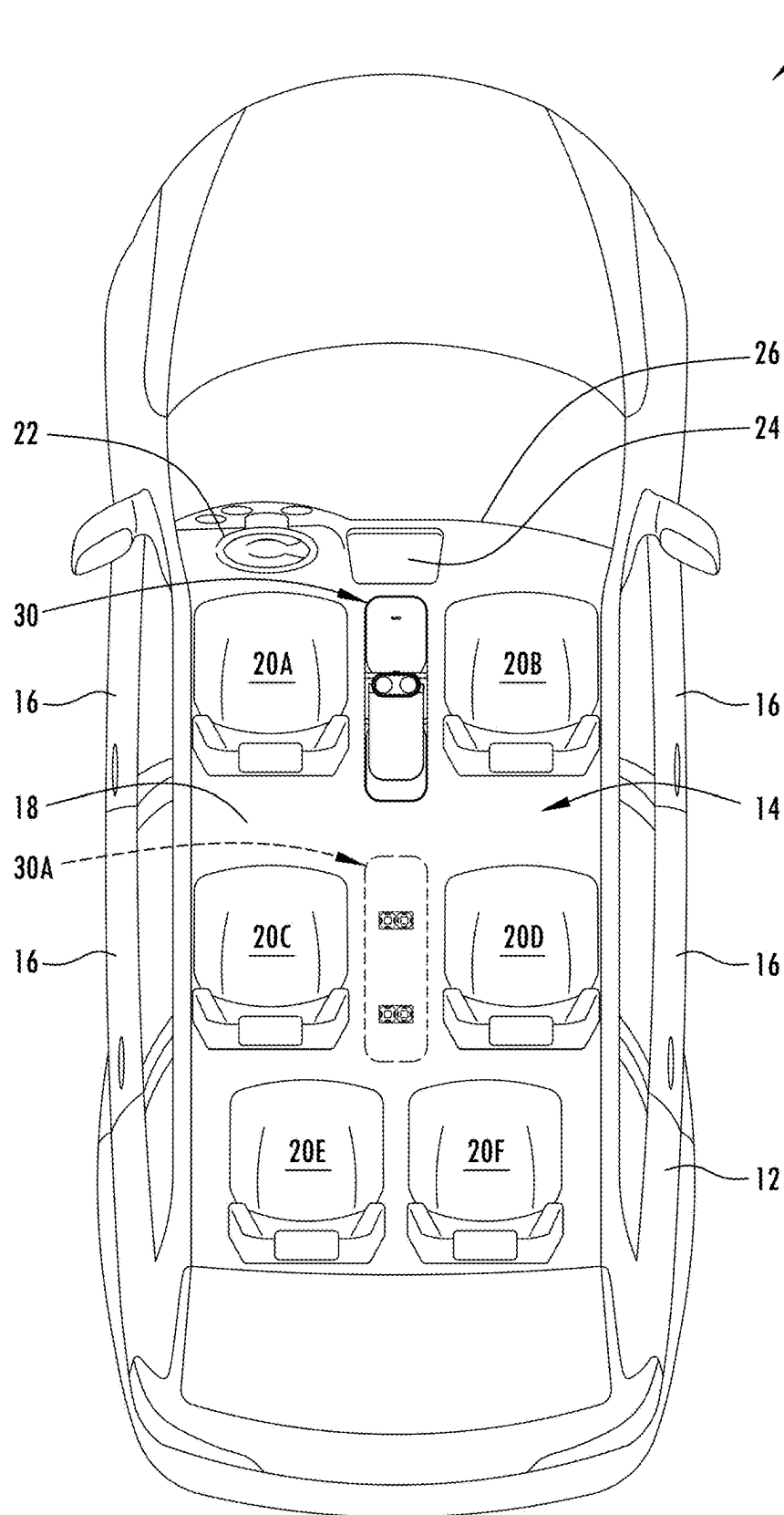


FIG. 1

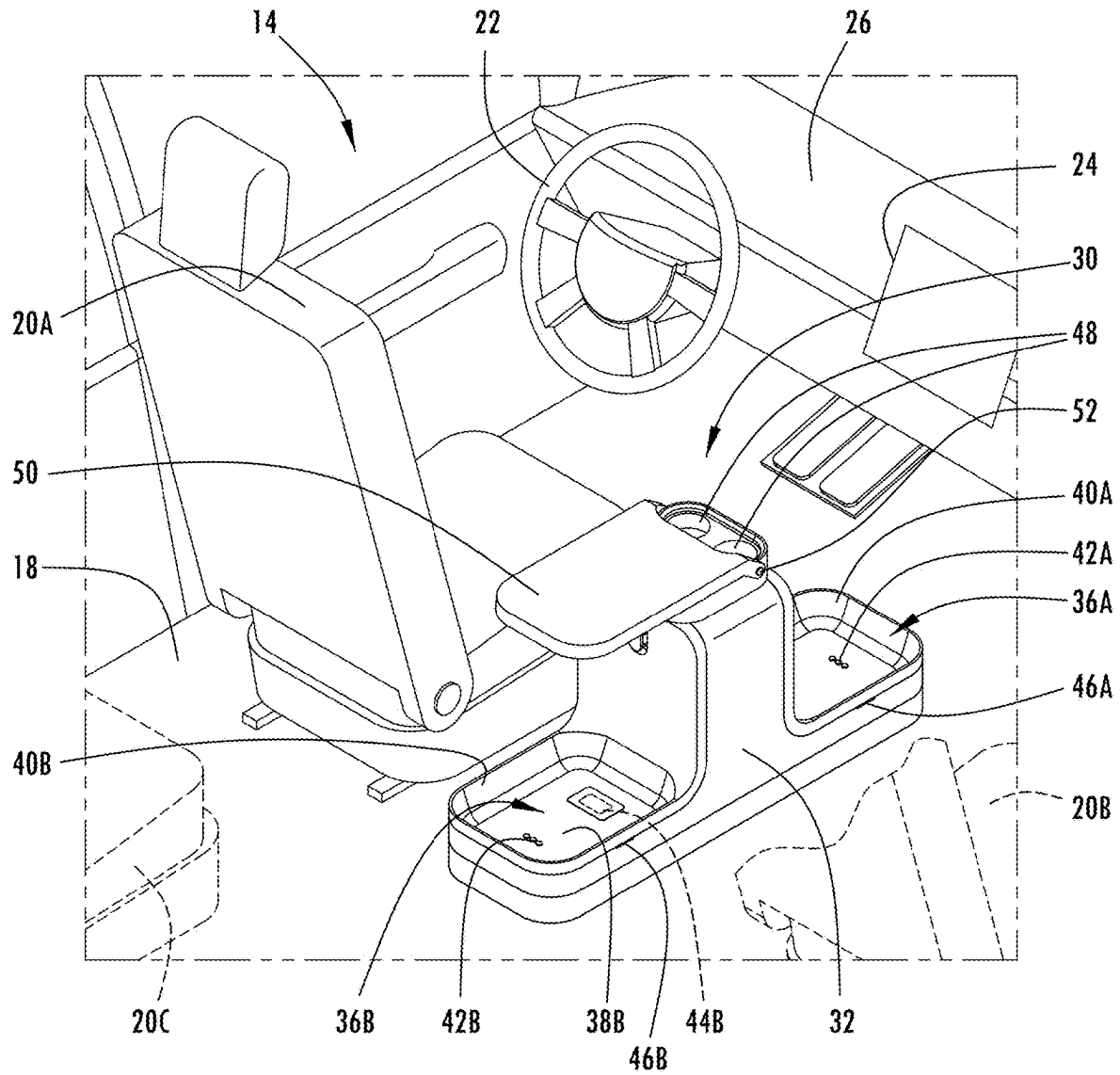


FIG. 2A

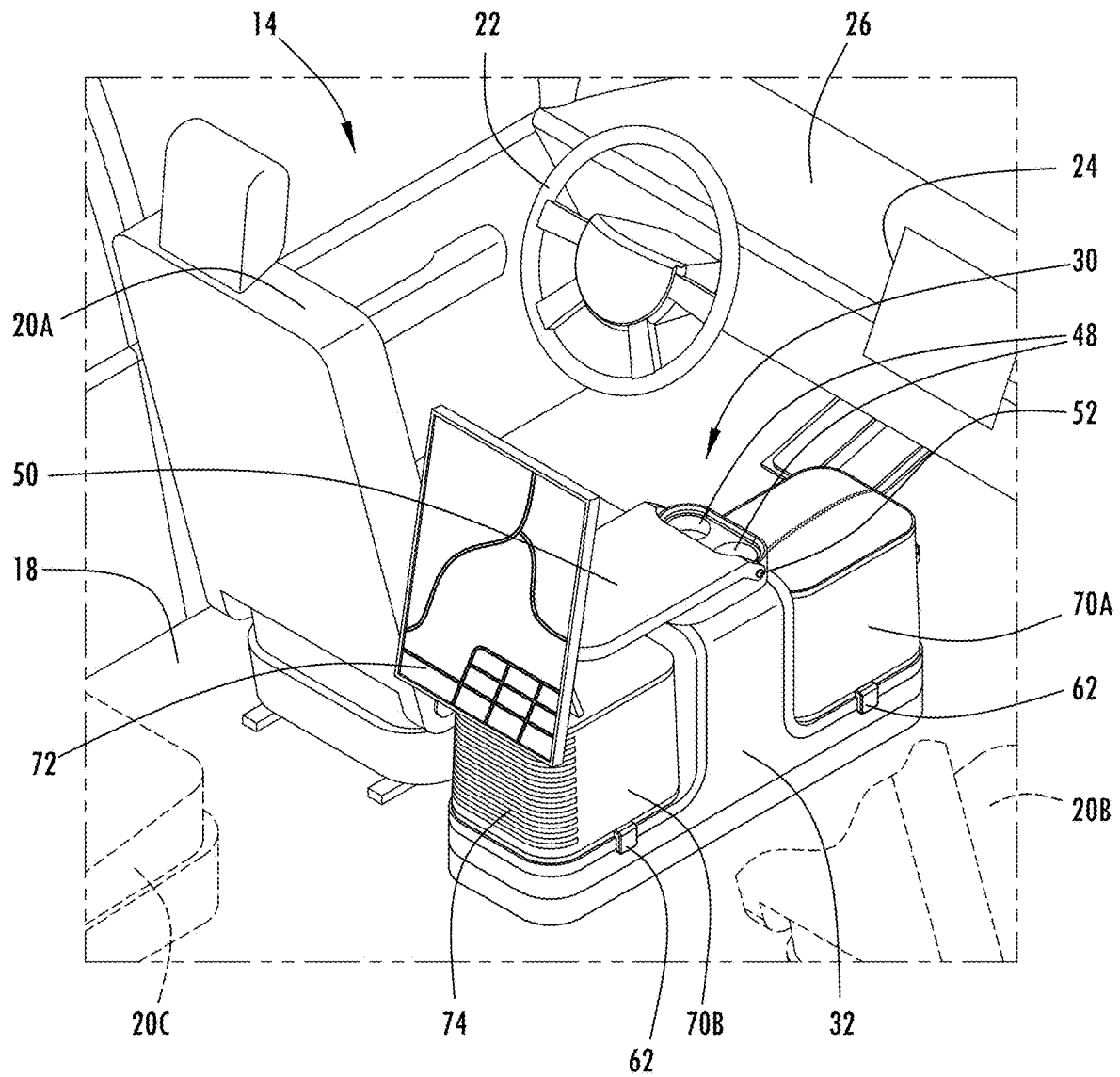


FIG. 2B

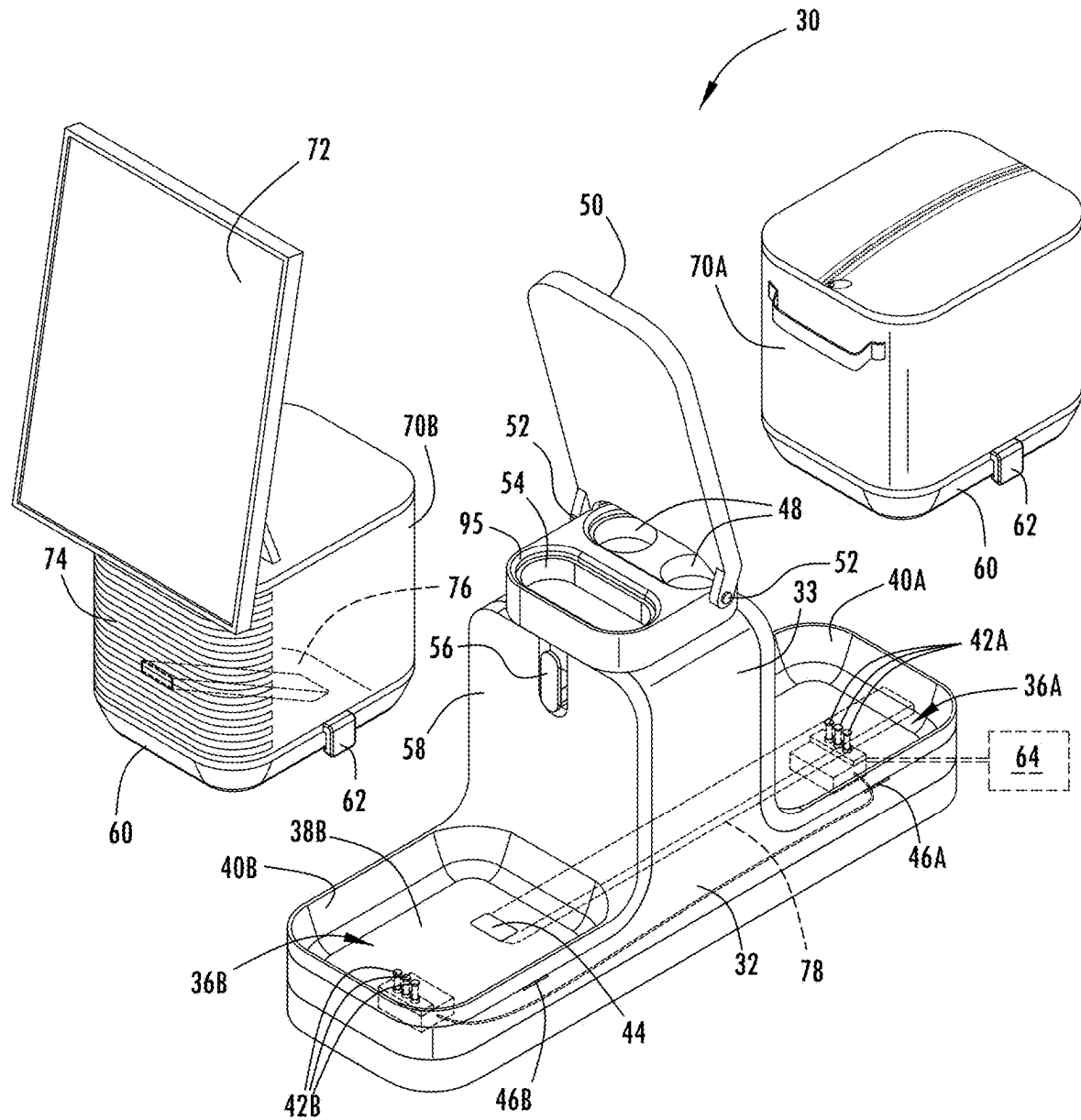


FIG. 3

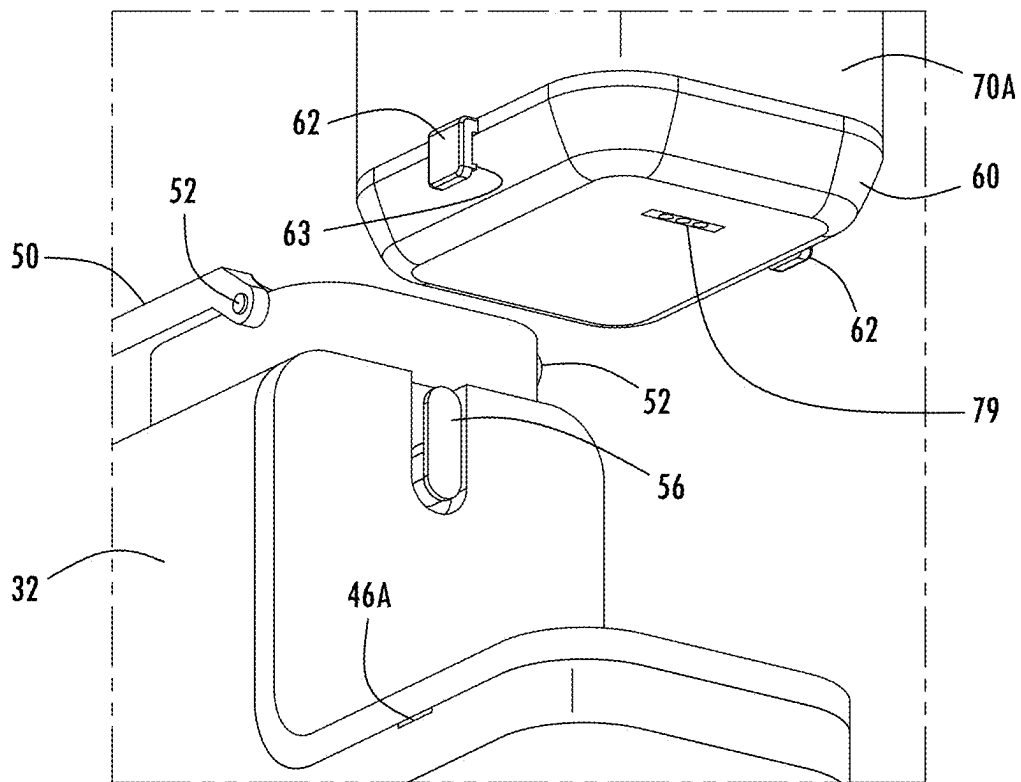


FIG. 4A

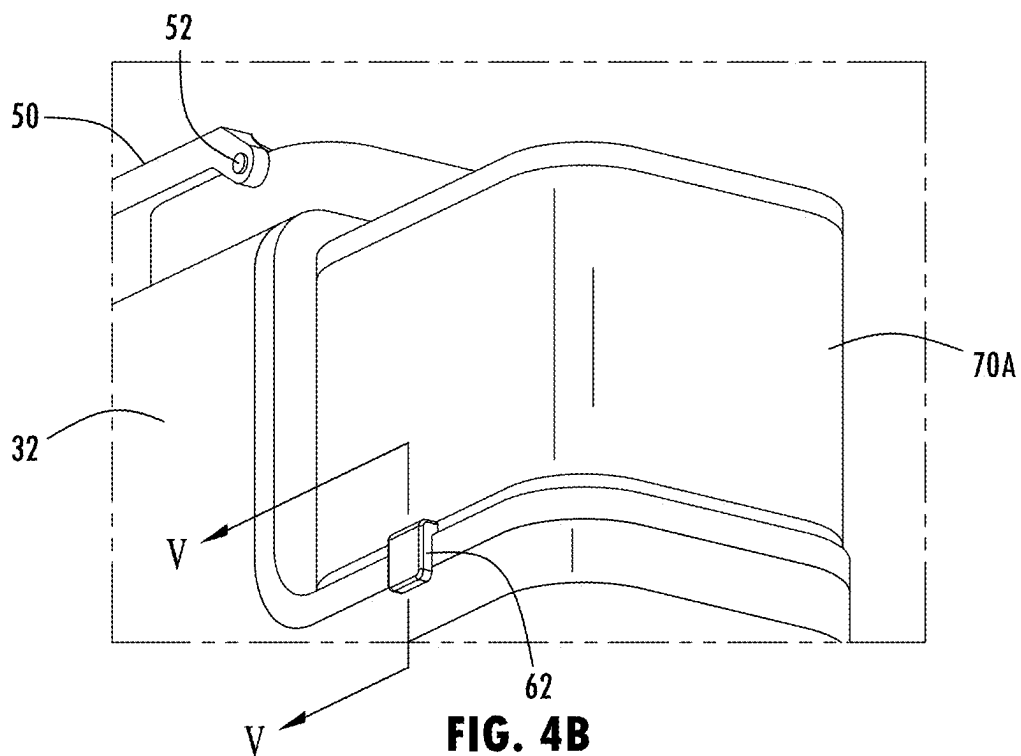


FIG. 4B

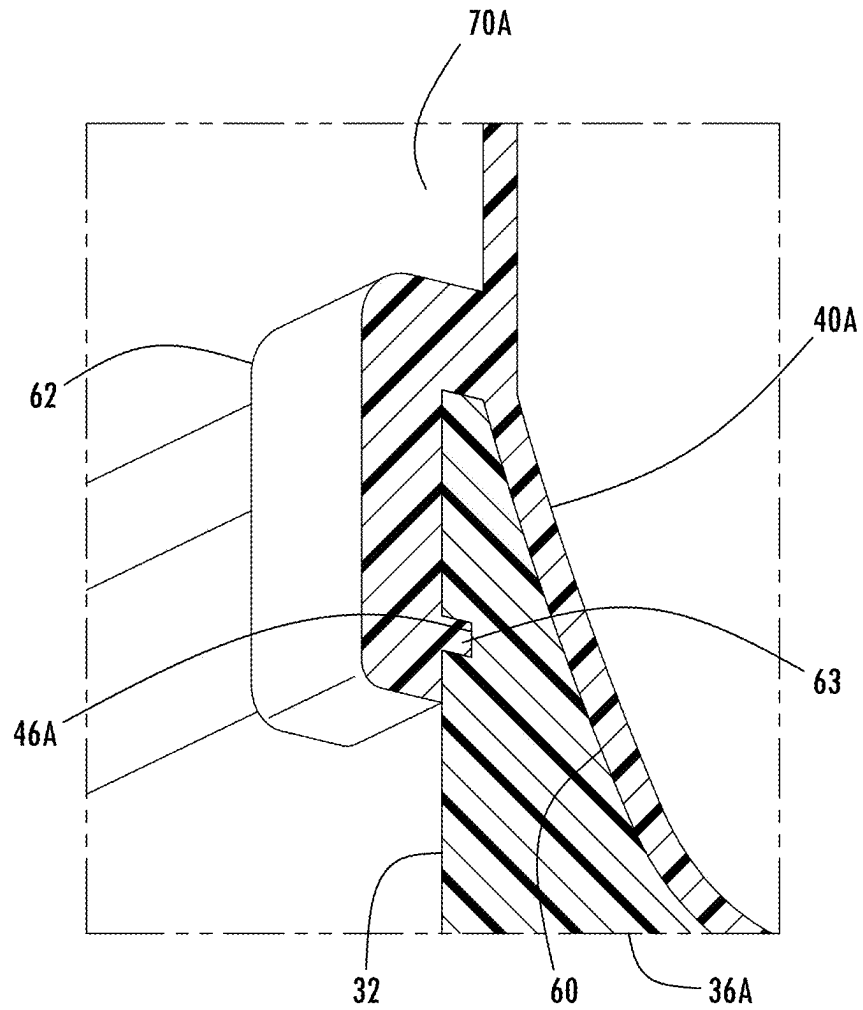


FIG. 5

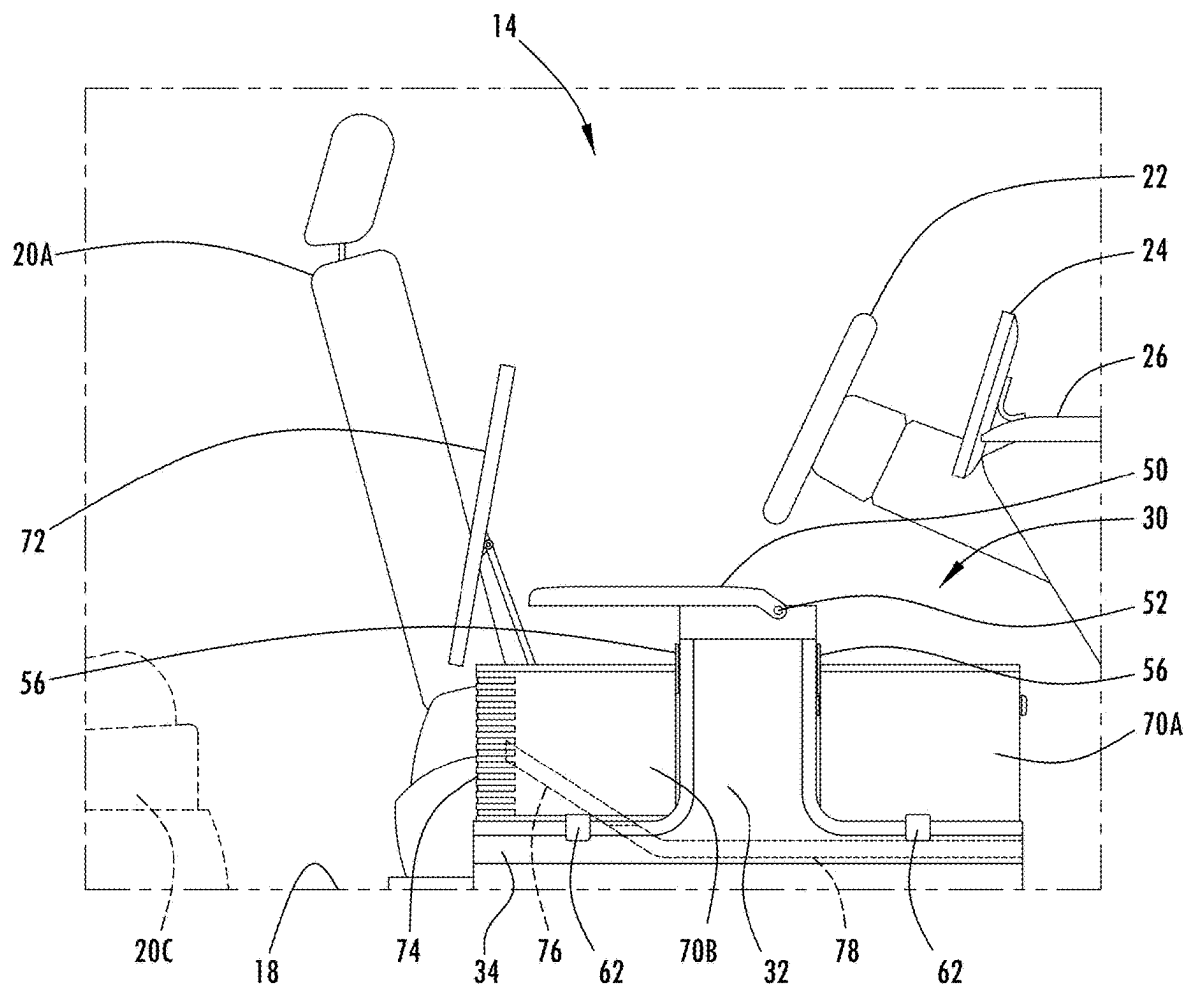


FIG. 6

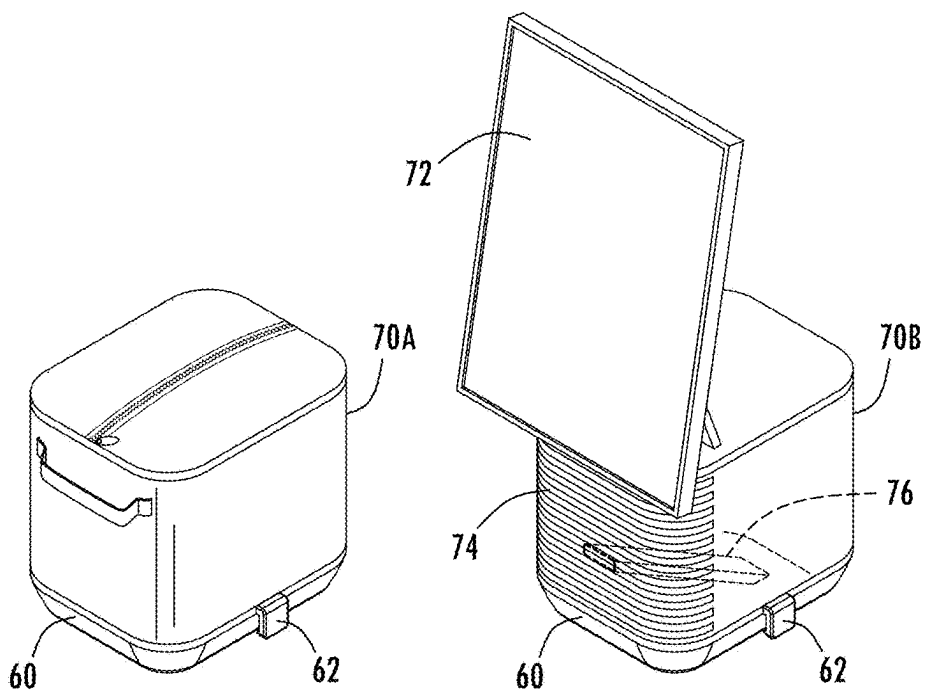


FIG. 7A

FIG. 7B

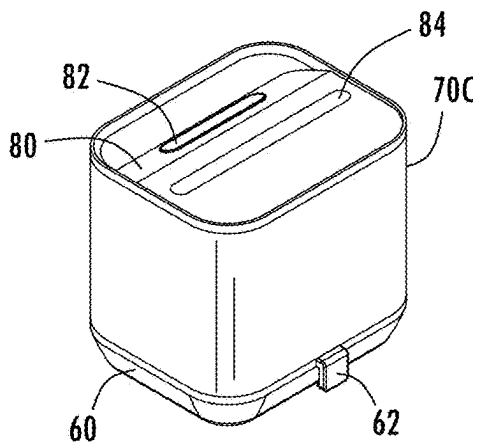


FIG. 7C

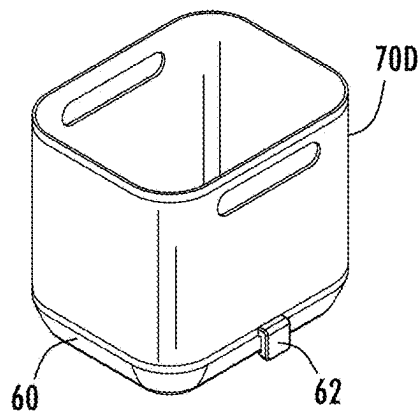


FIG. 7D

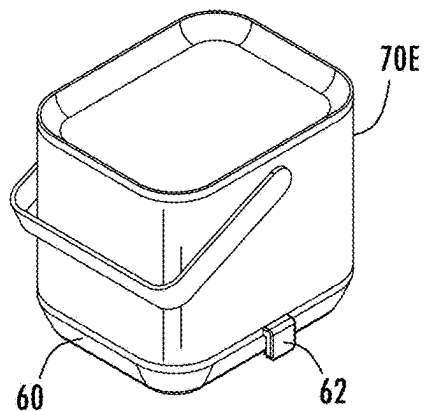


FIG. 7E

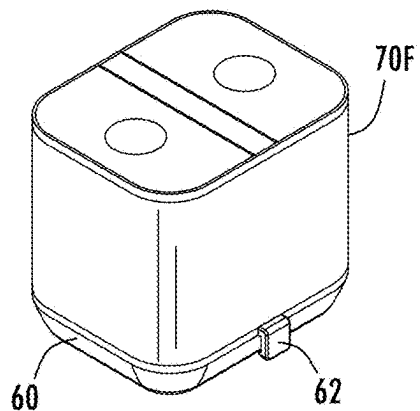


FIG. 7F

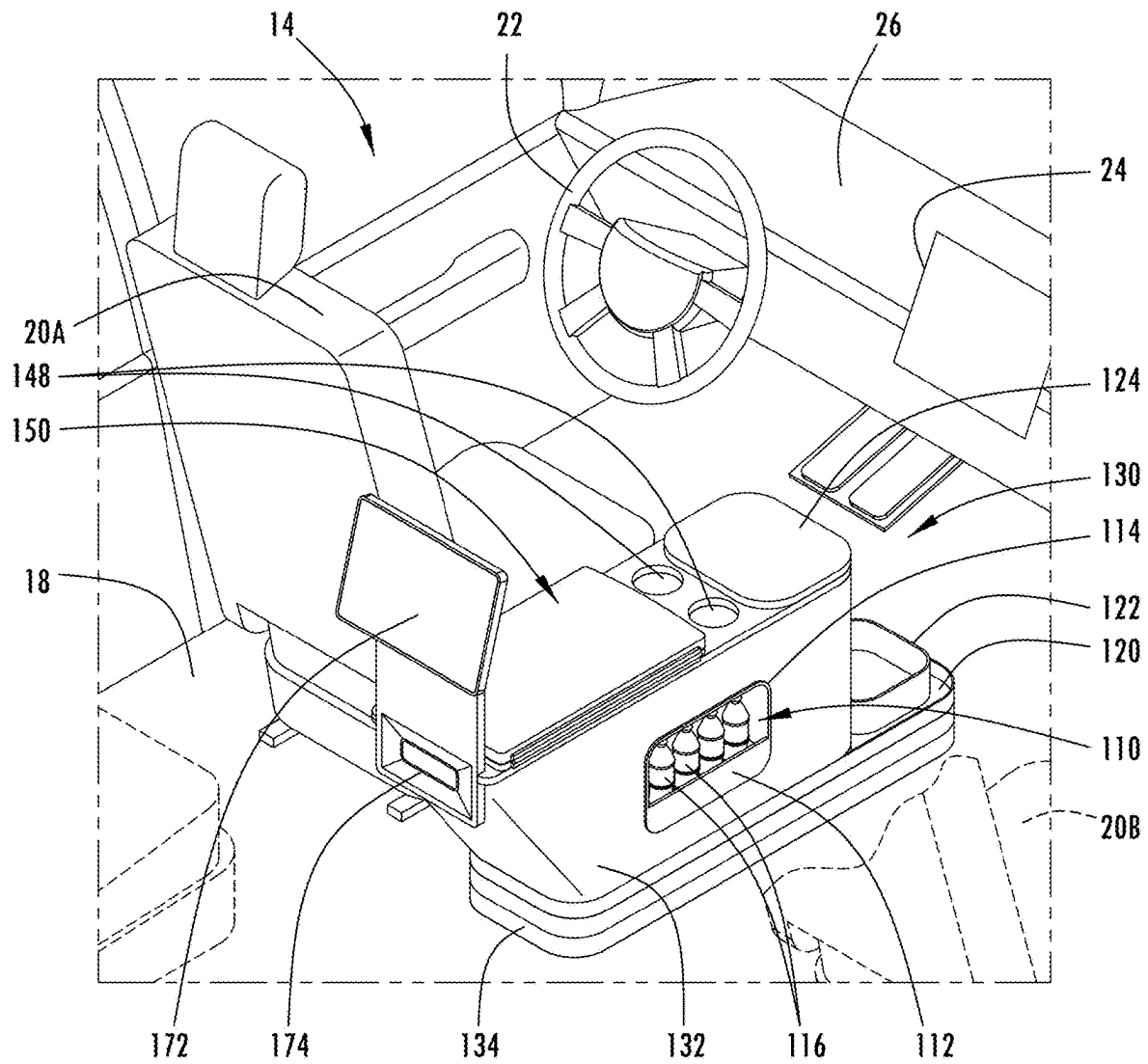


FIG. 8A

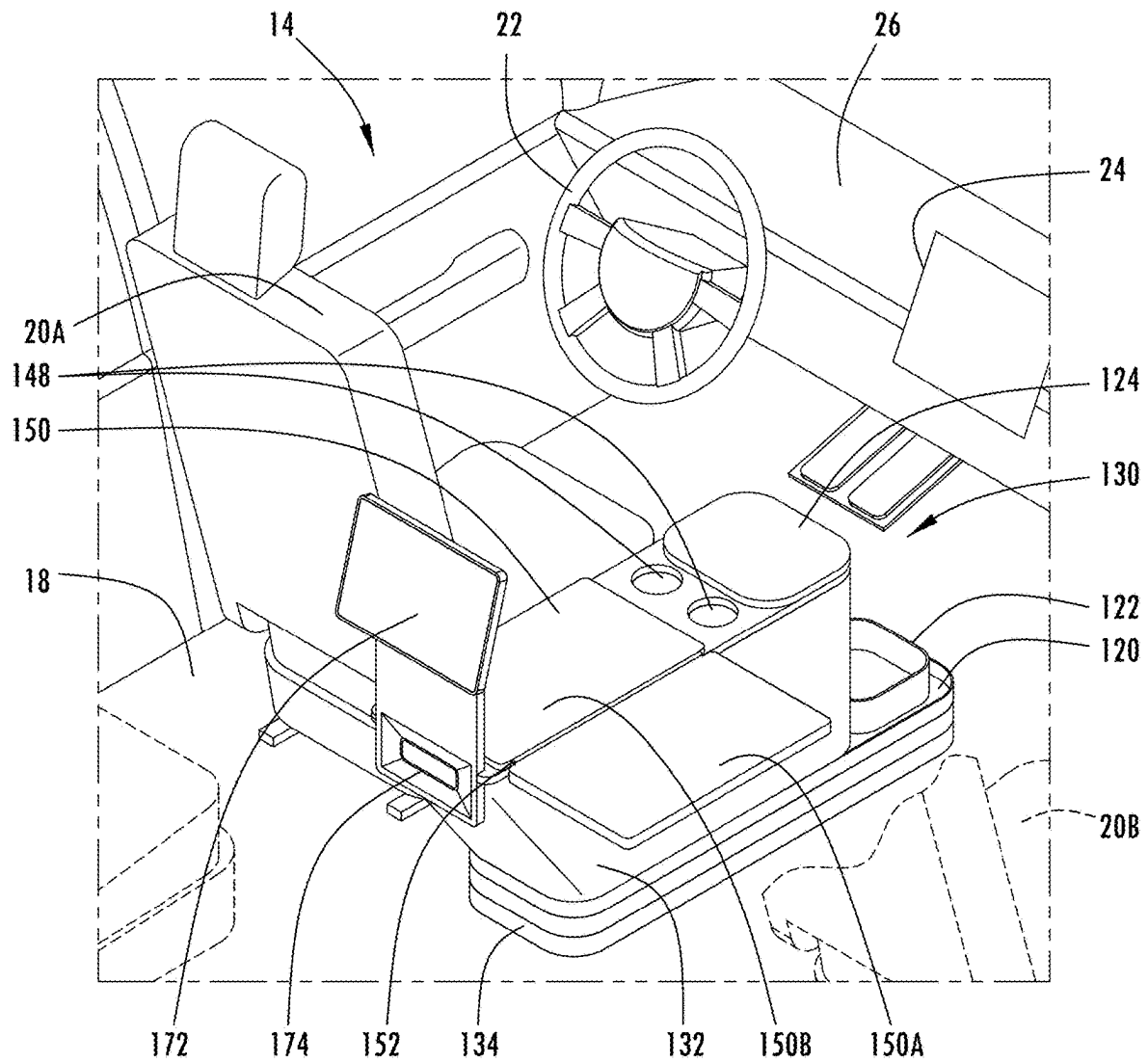


FIG. 8B

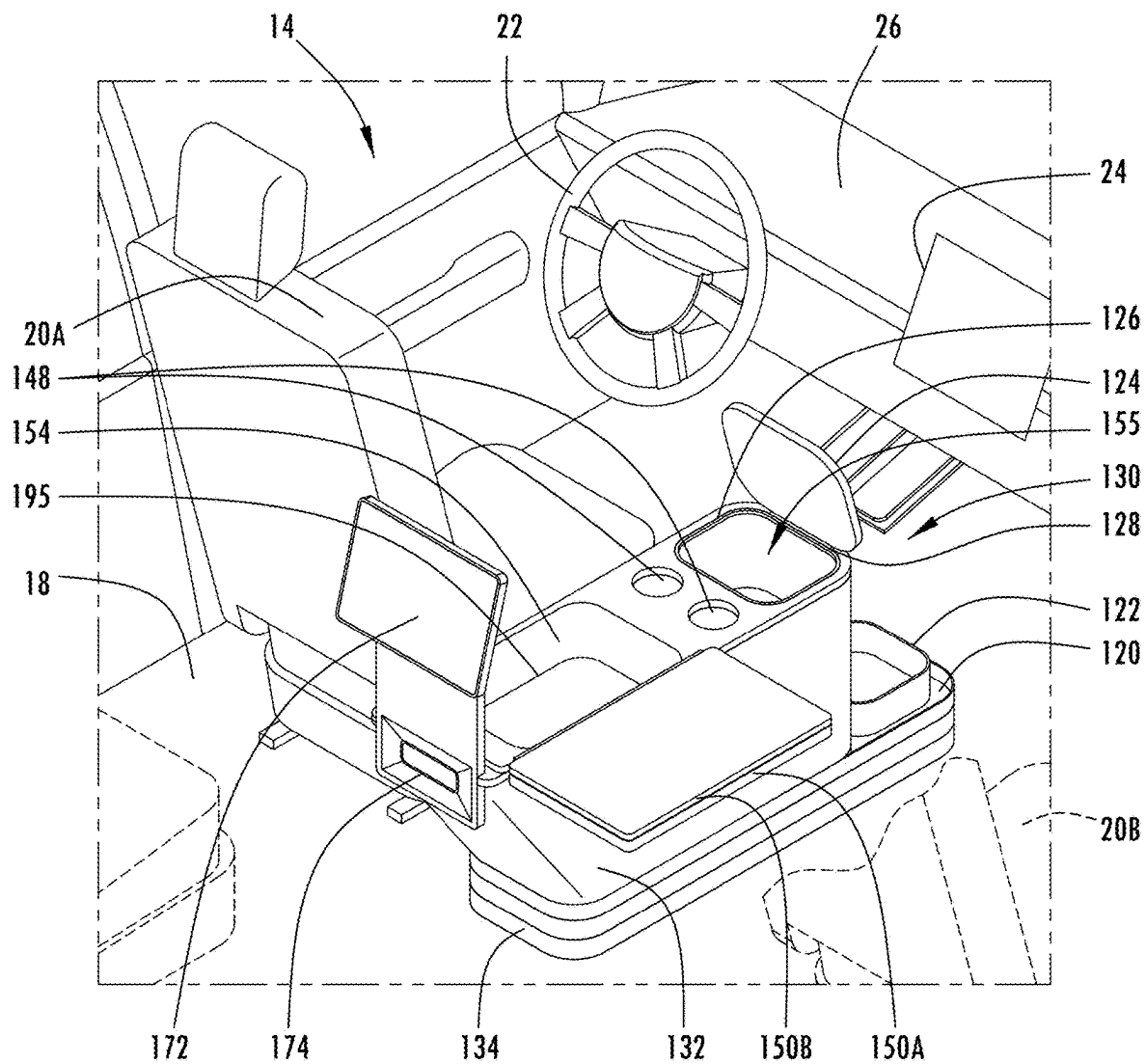


FIG. 8C

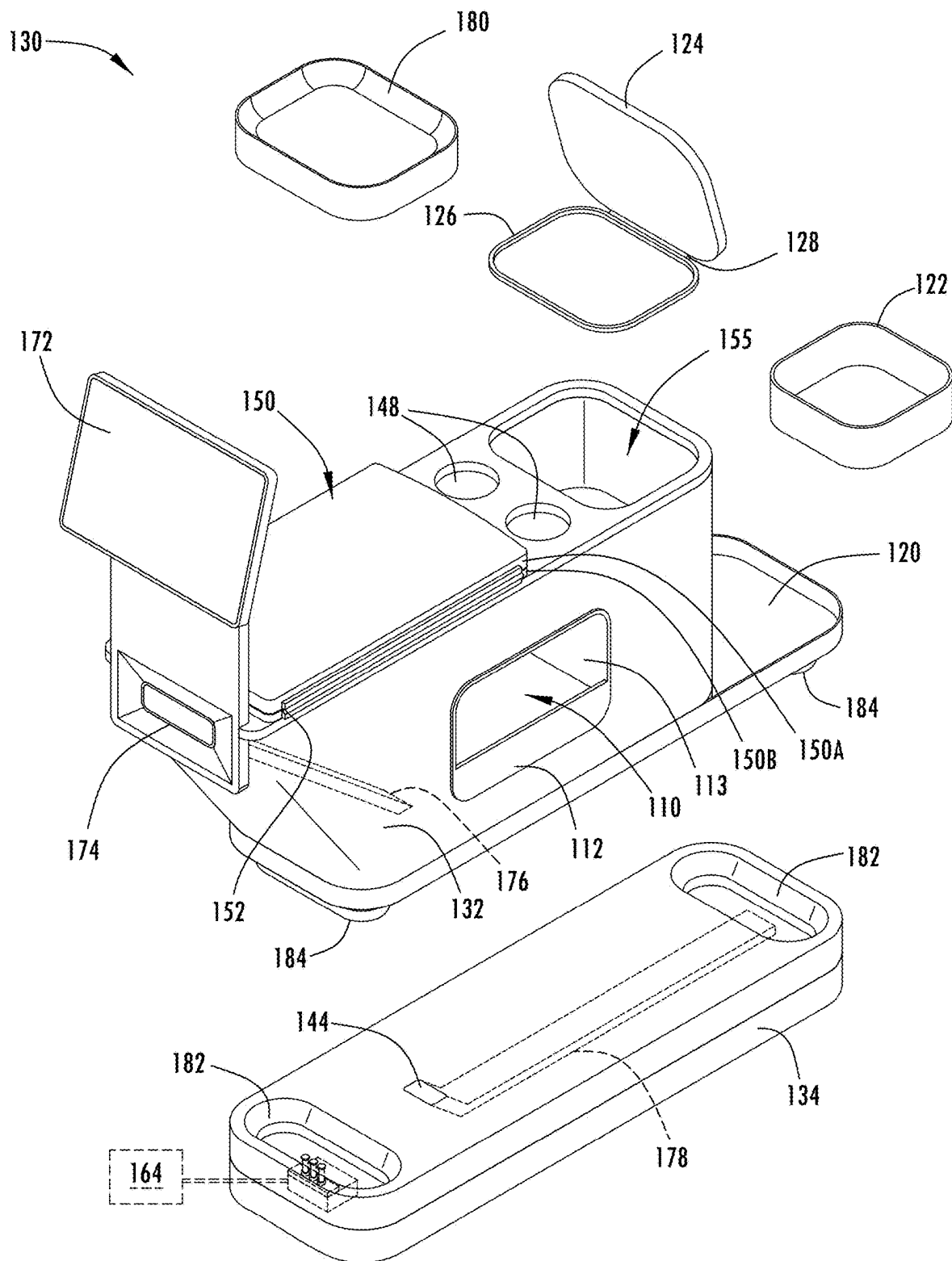


FIG. 9

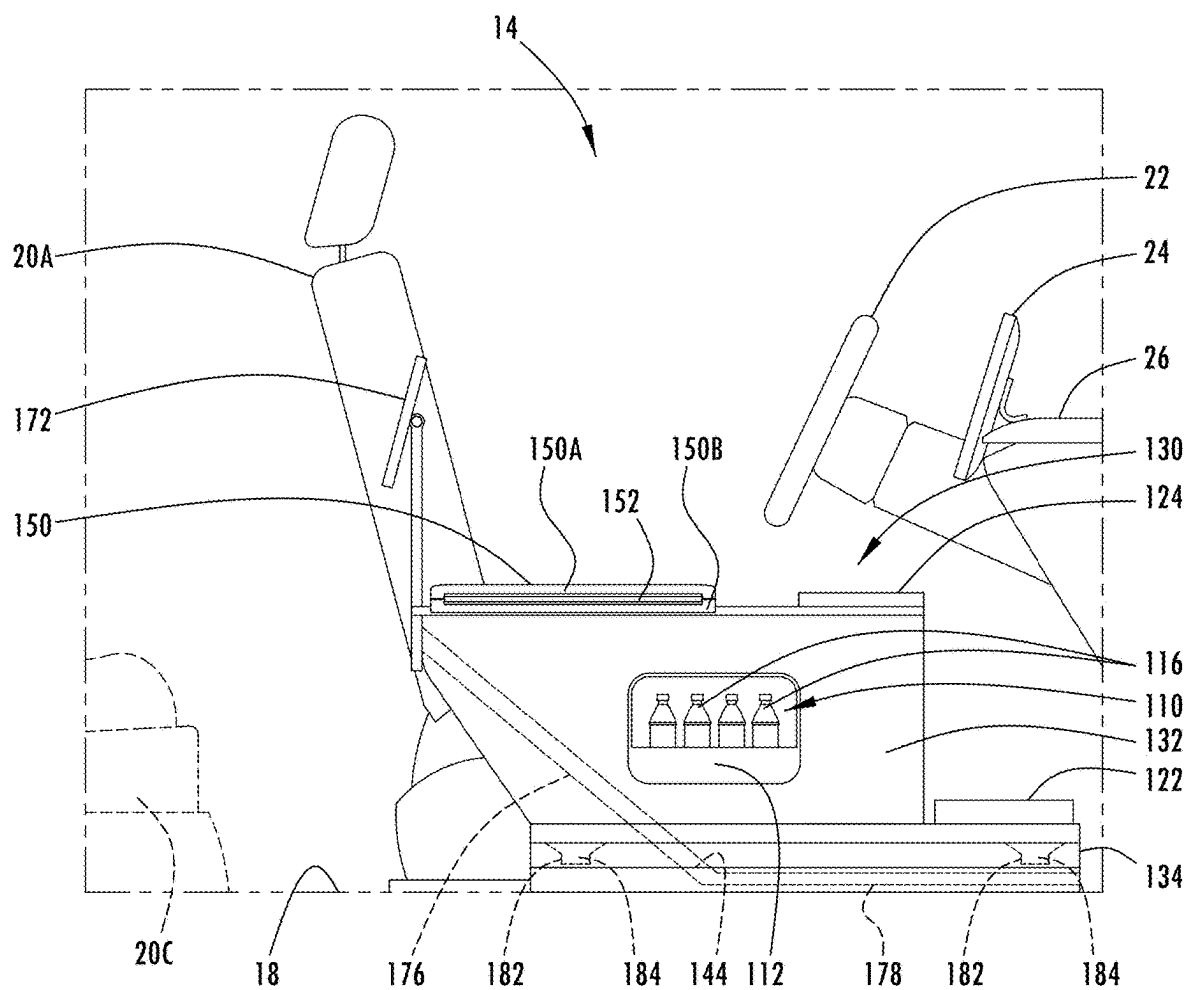


FIG. 10

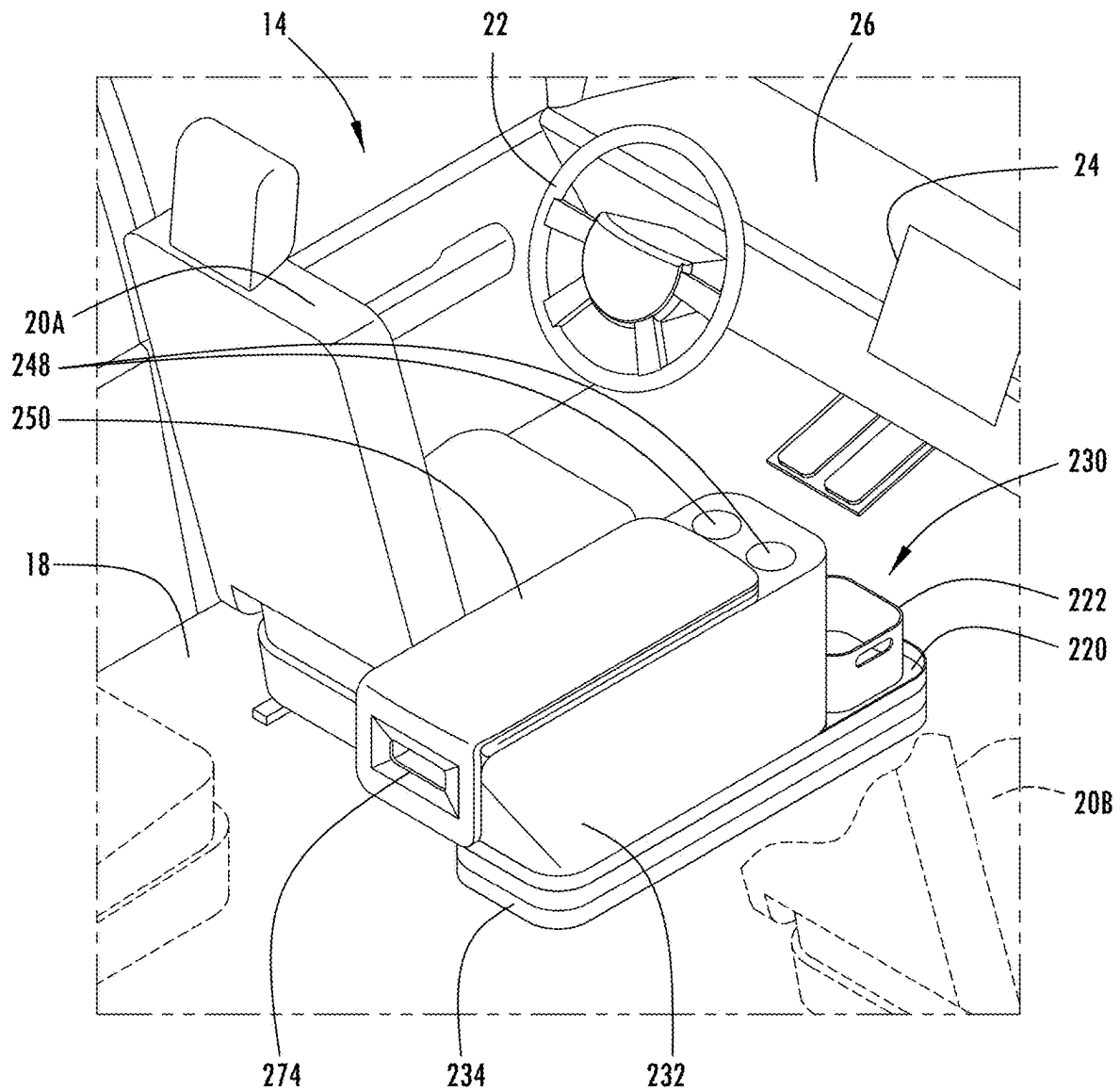


FIG. 11A

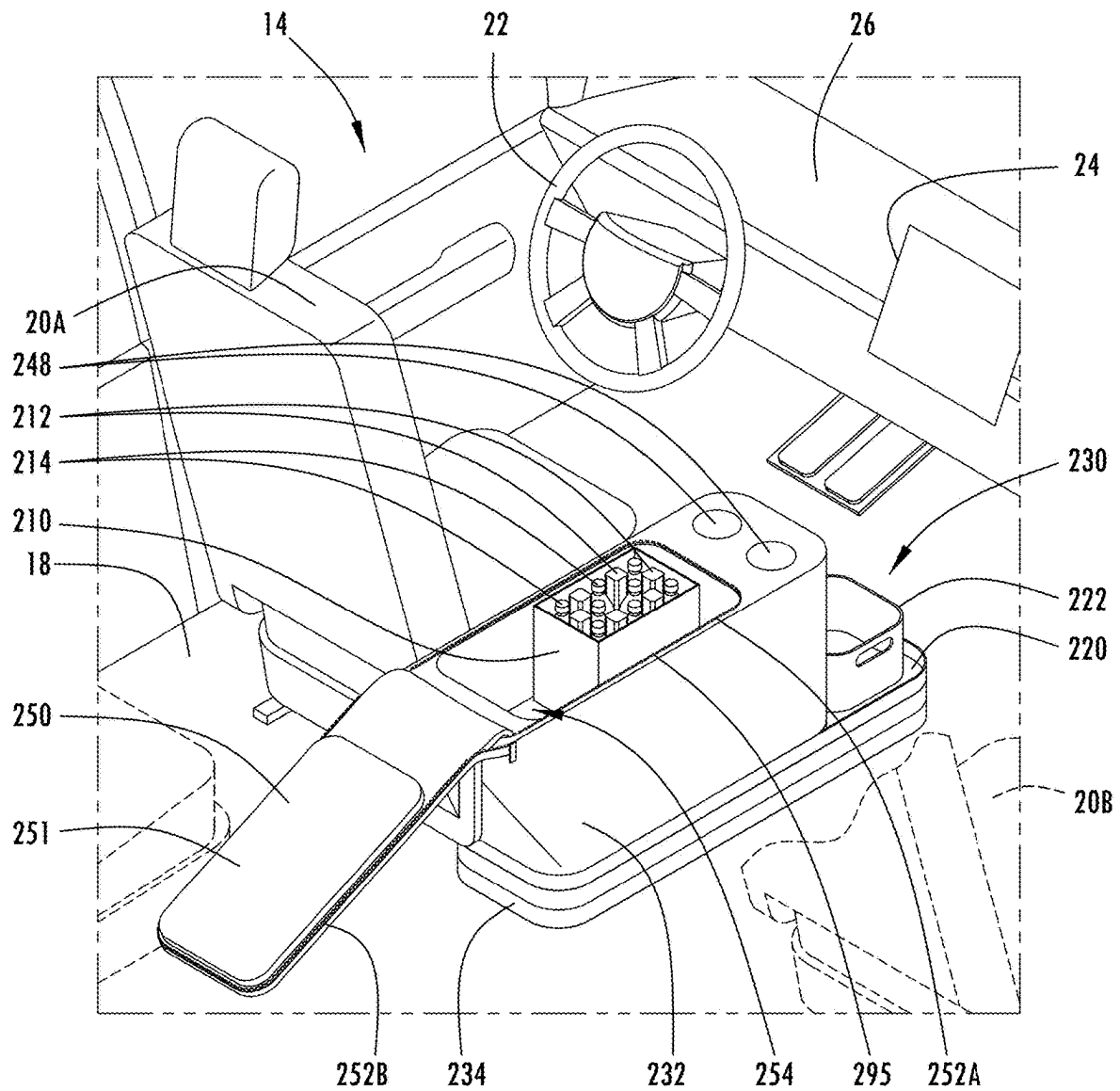


FIG. 11B

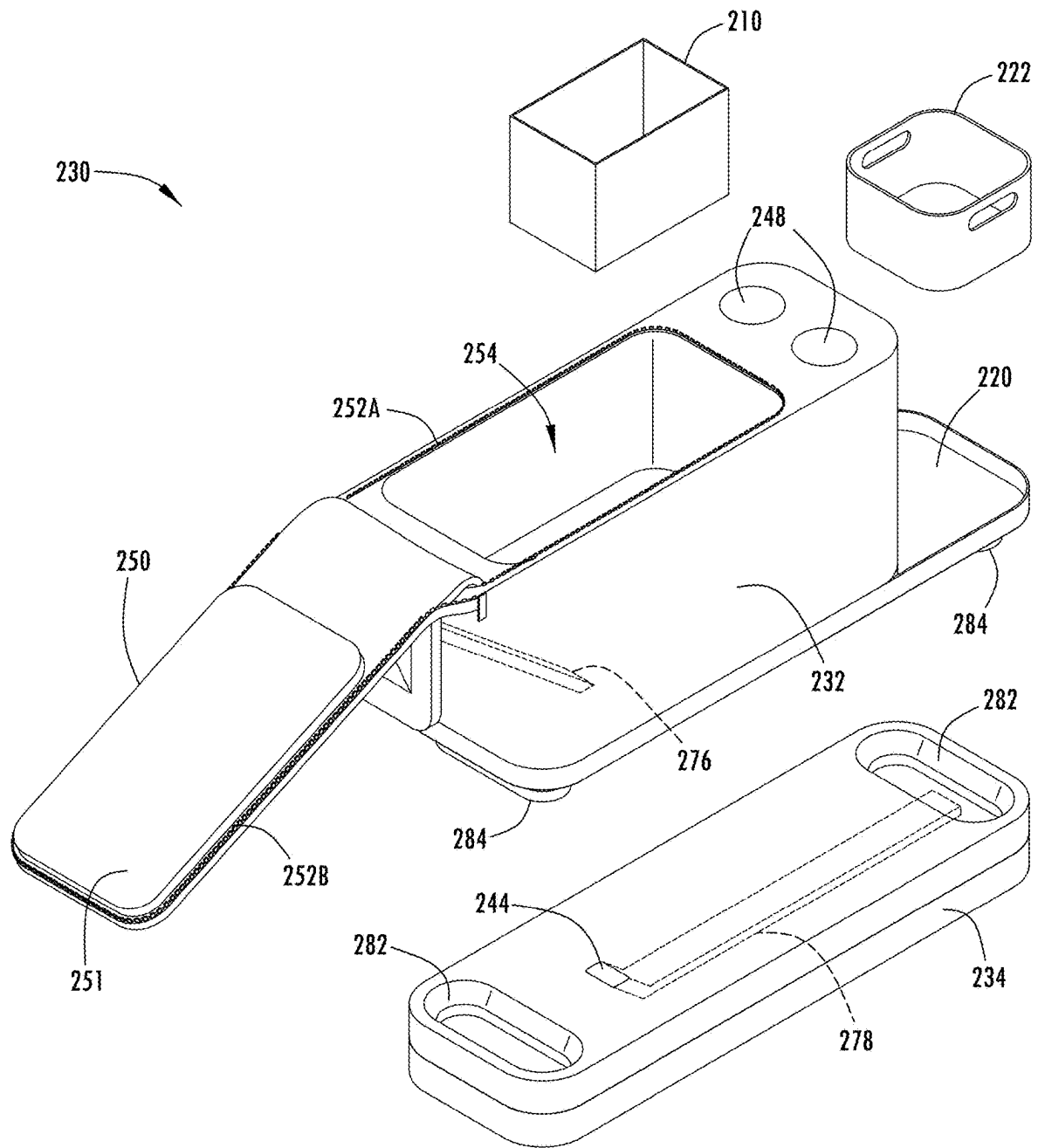


FIG. 12

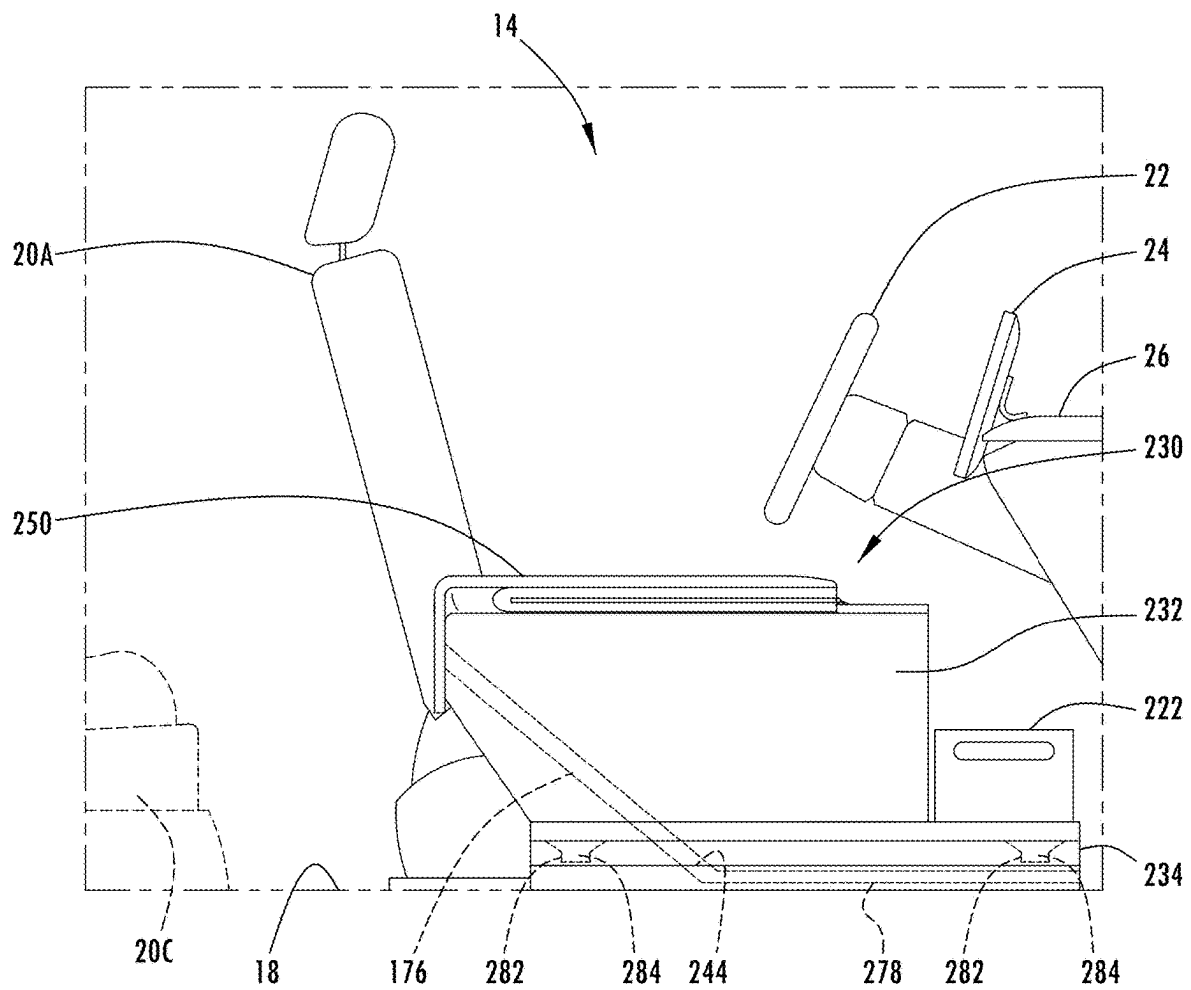


FIG. 13

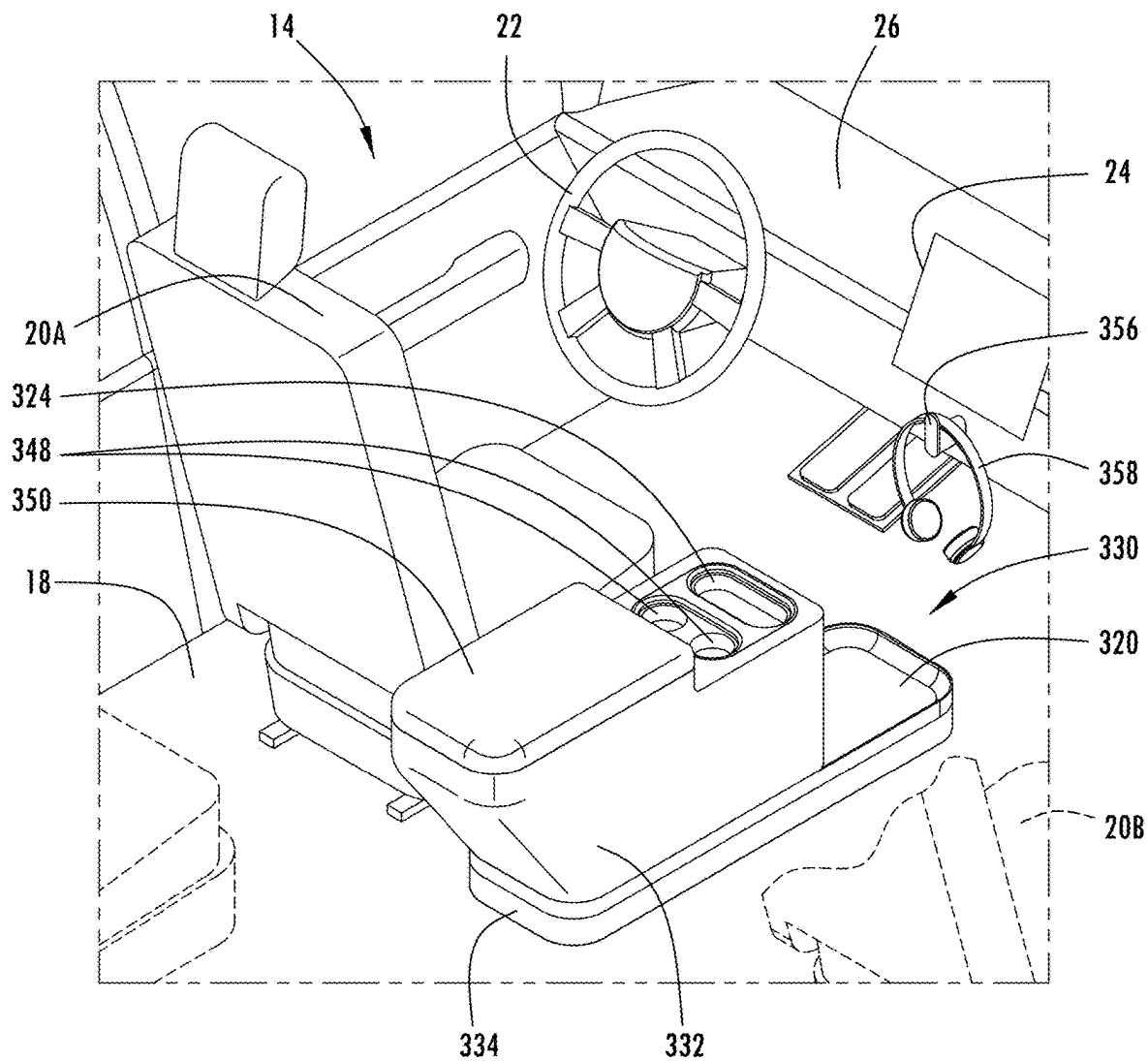


FIG. 14A

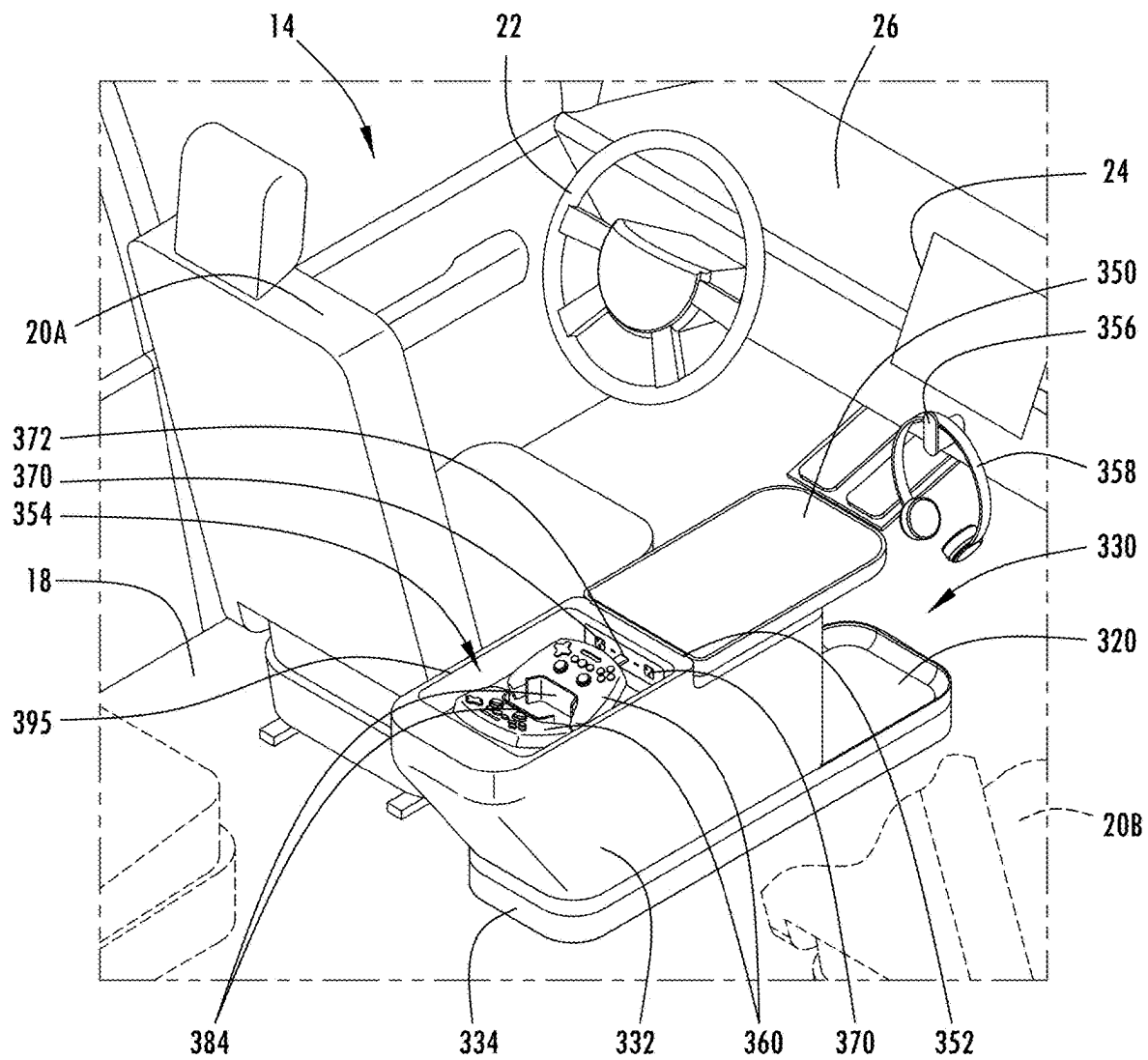


FIG. 14B

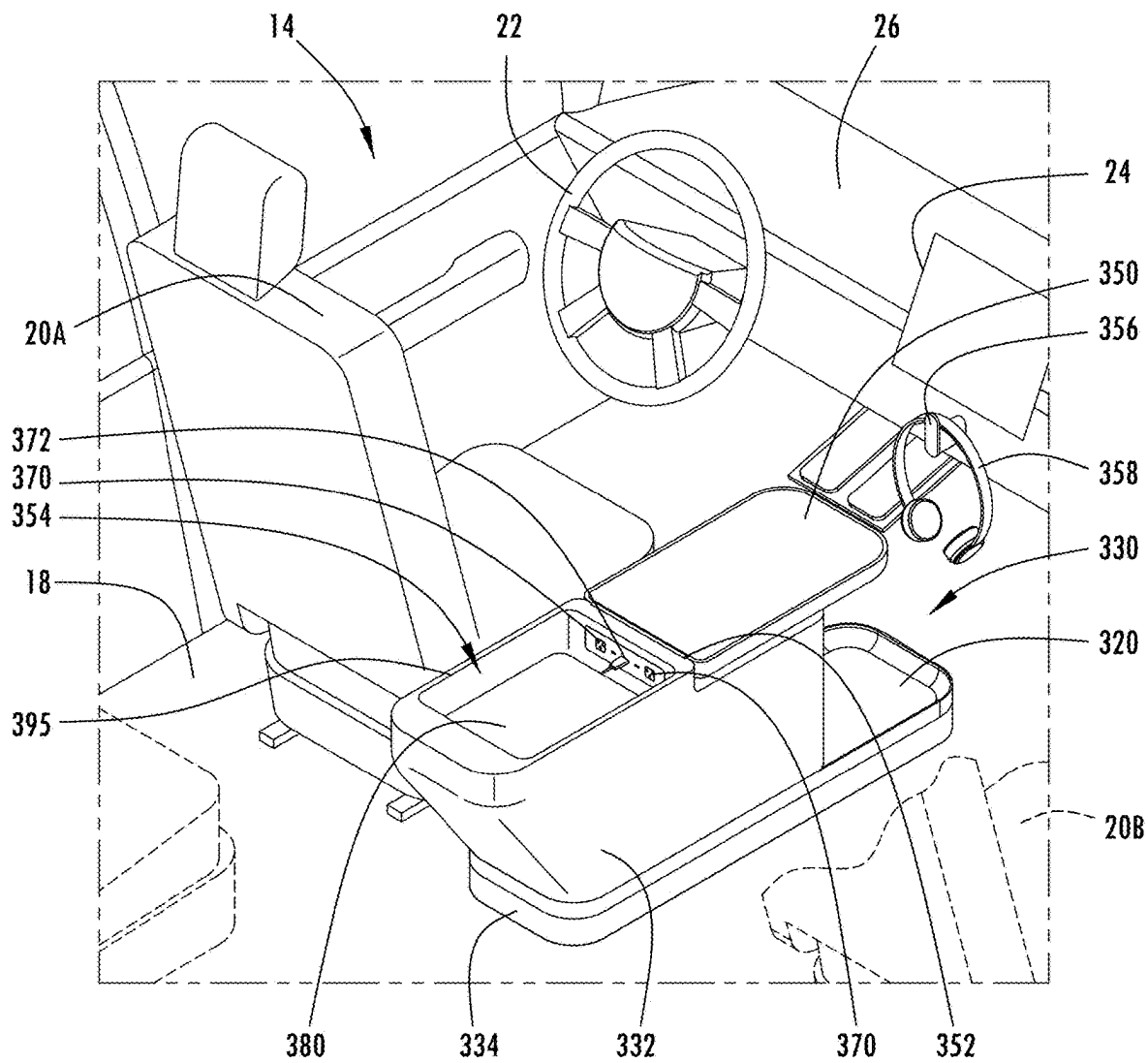


FIG. 14C

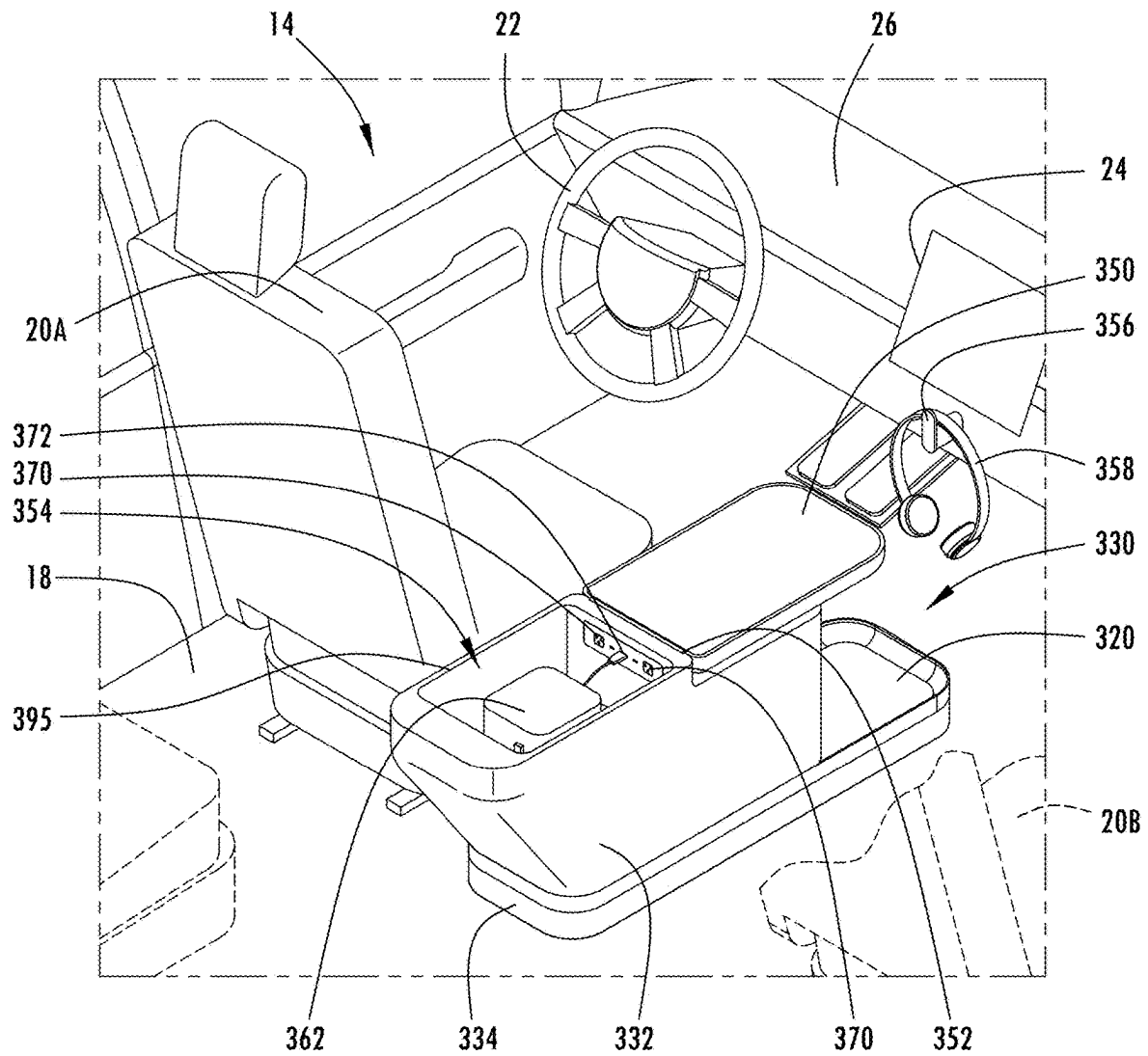


FIG. 14D

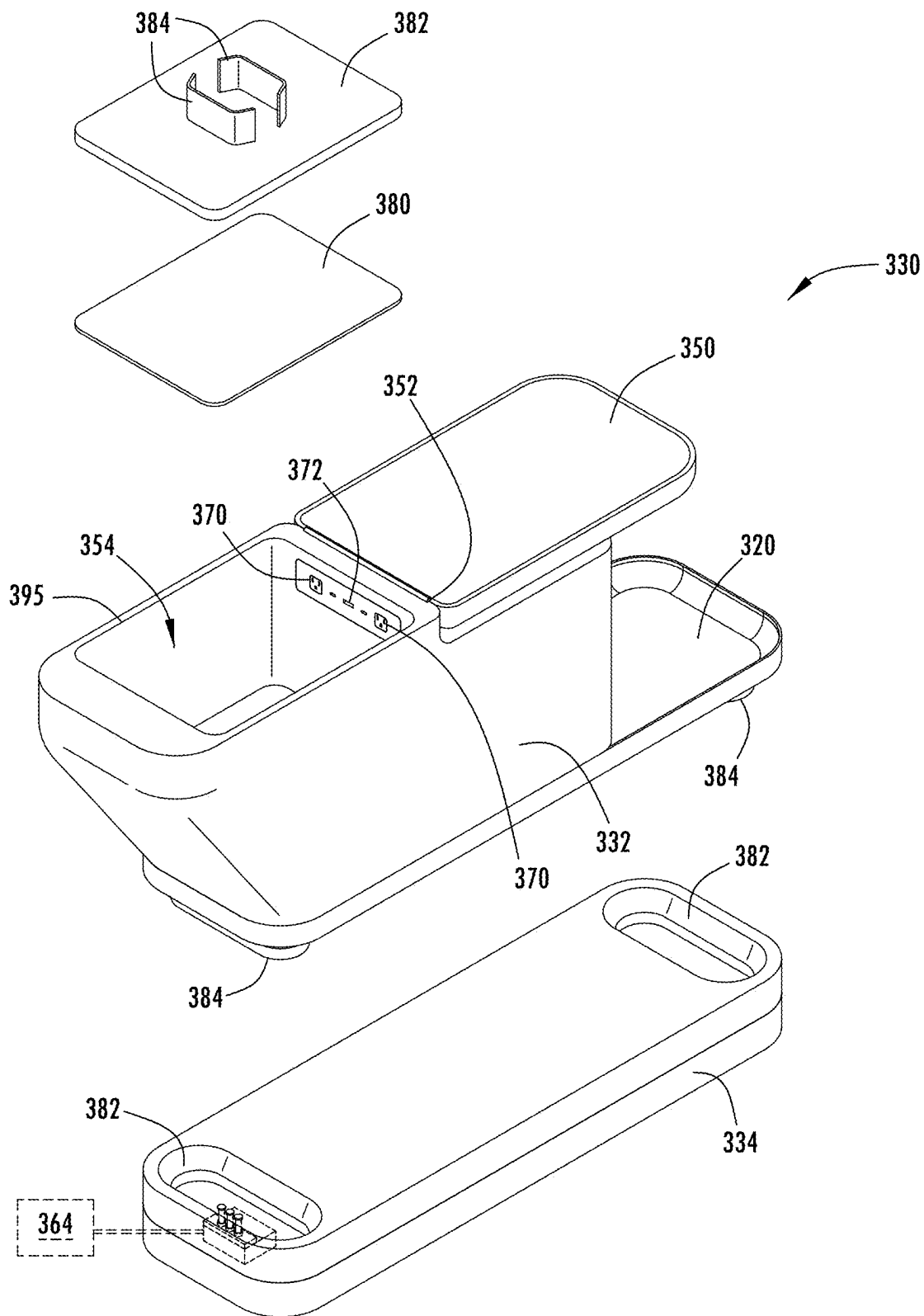


FIG. 15

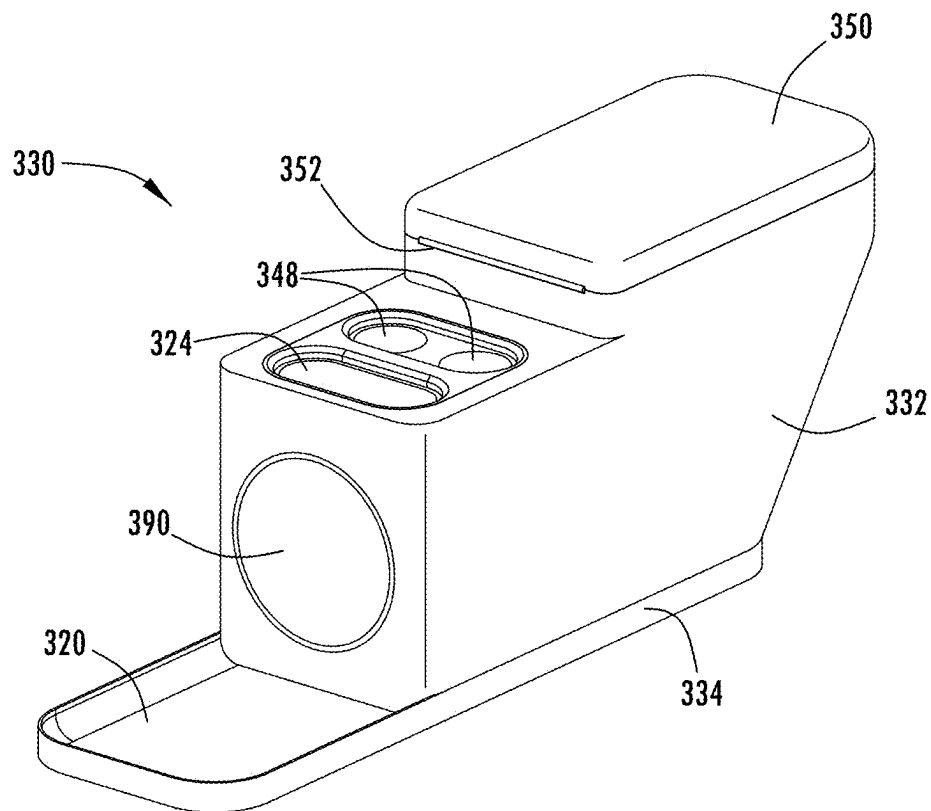


FIG. 16

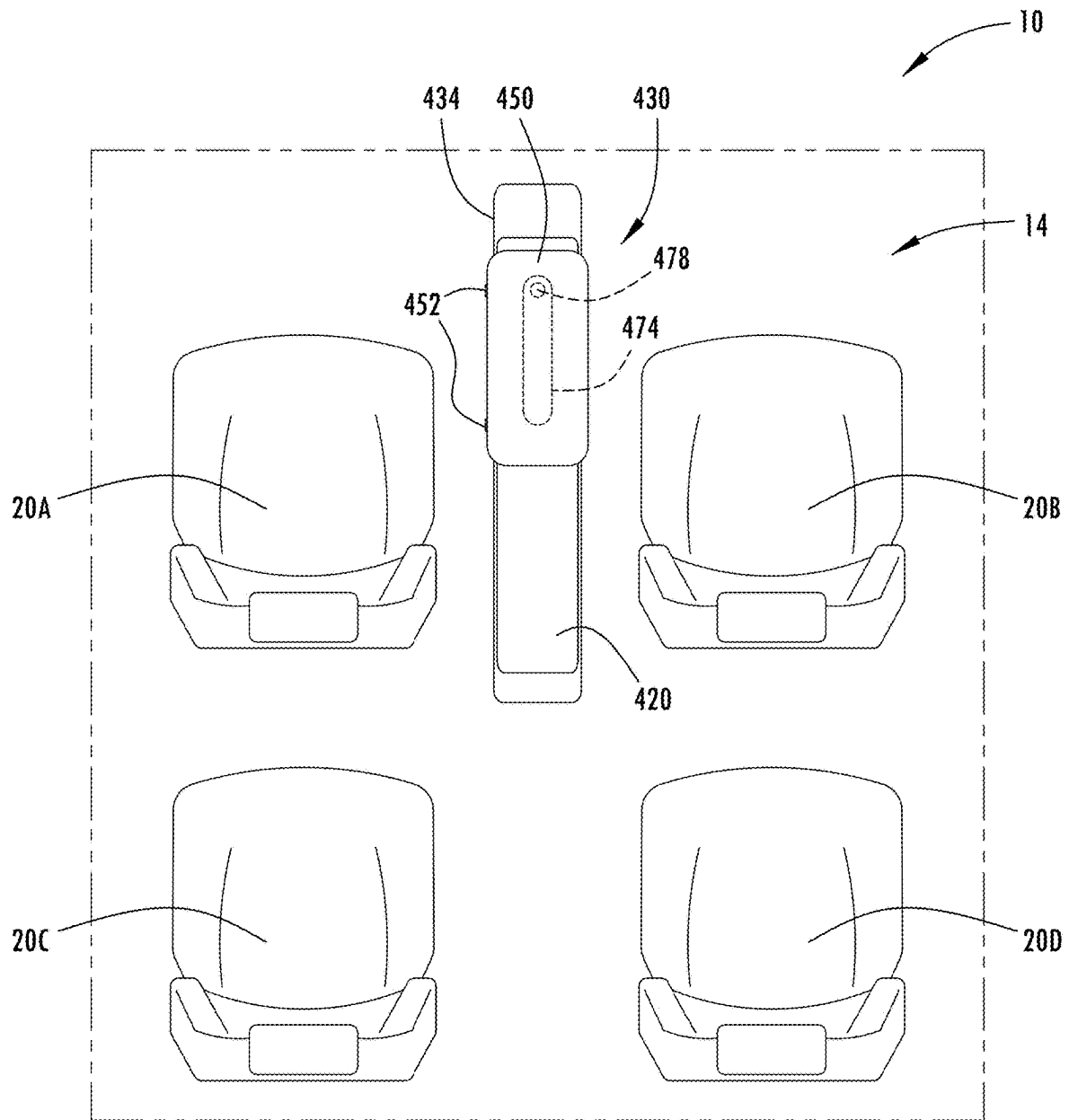
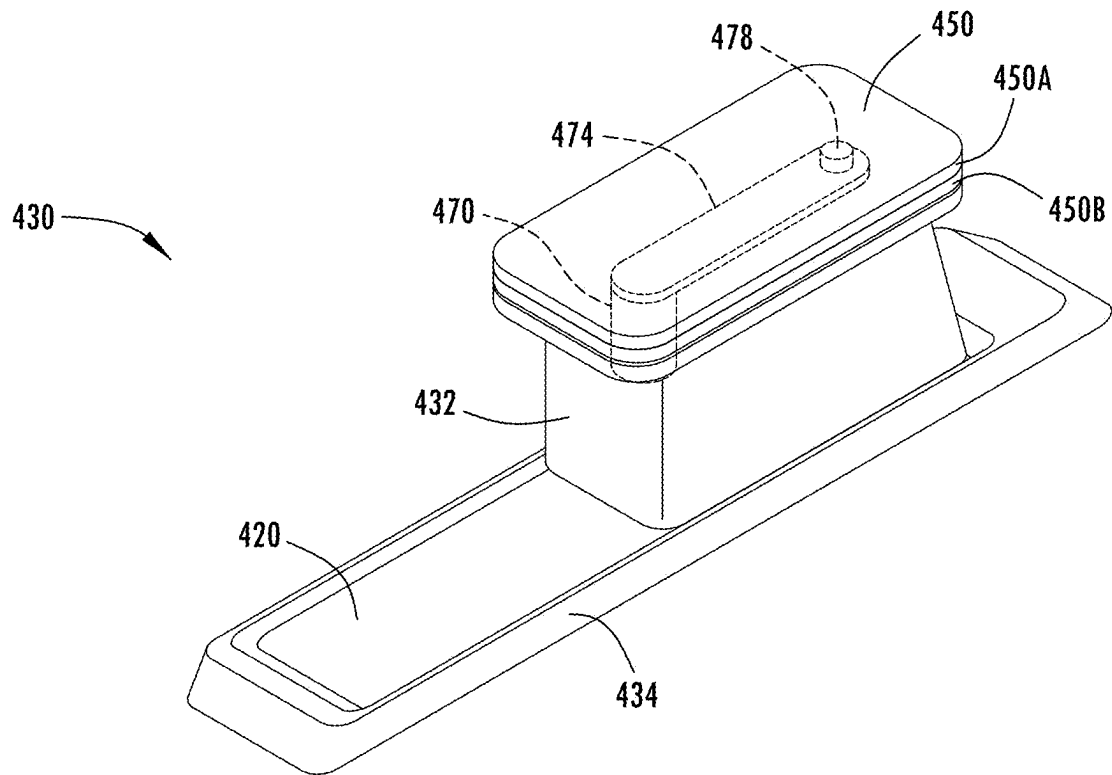


FIG. 18A

**FIG. 18B**

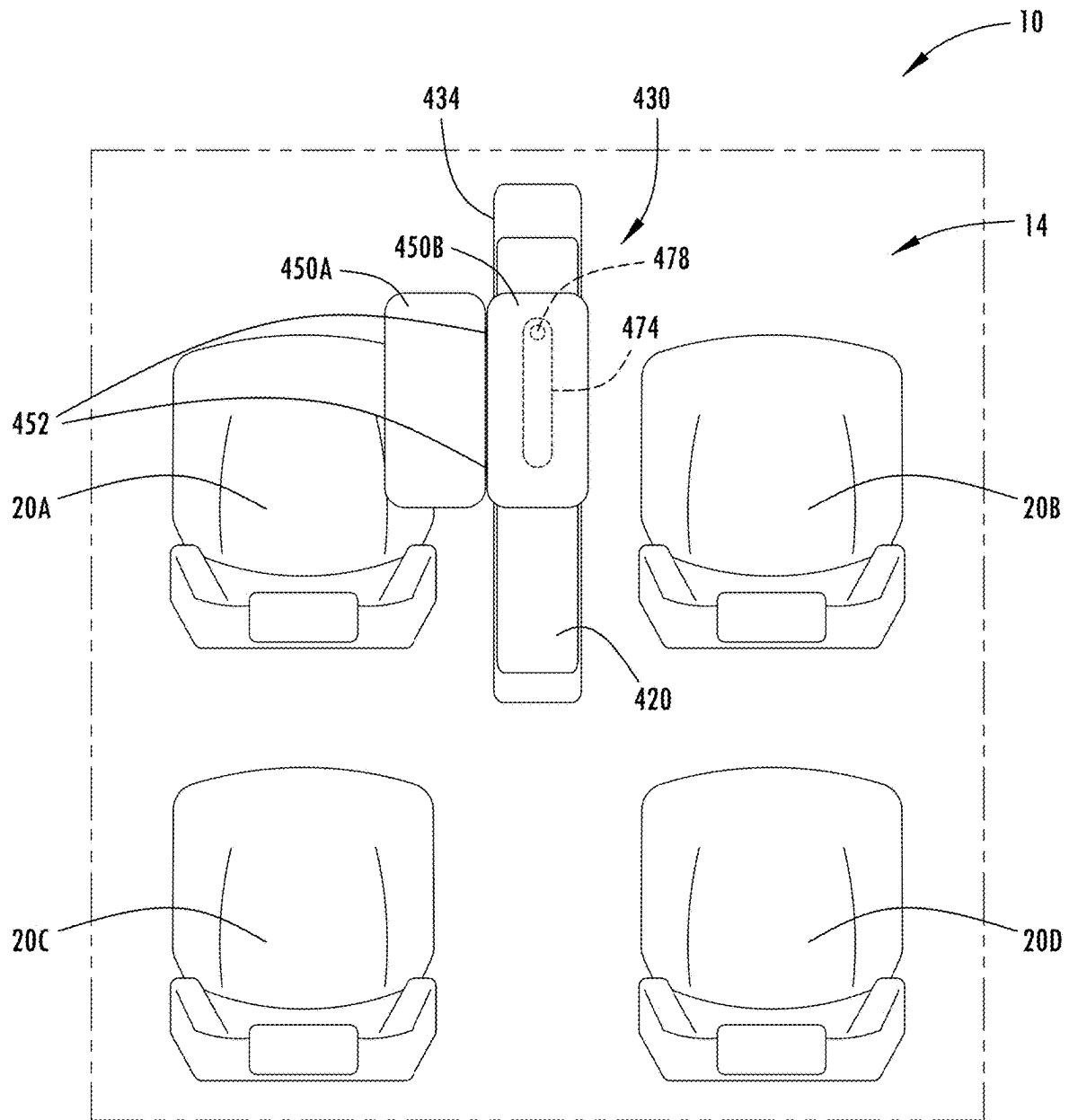


FIG. 19A

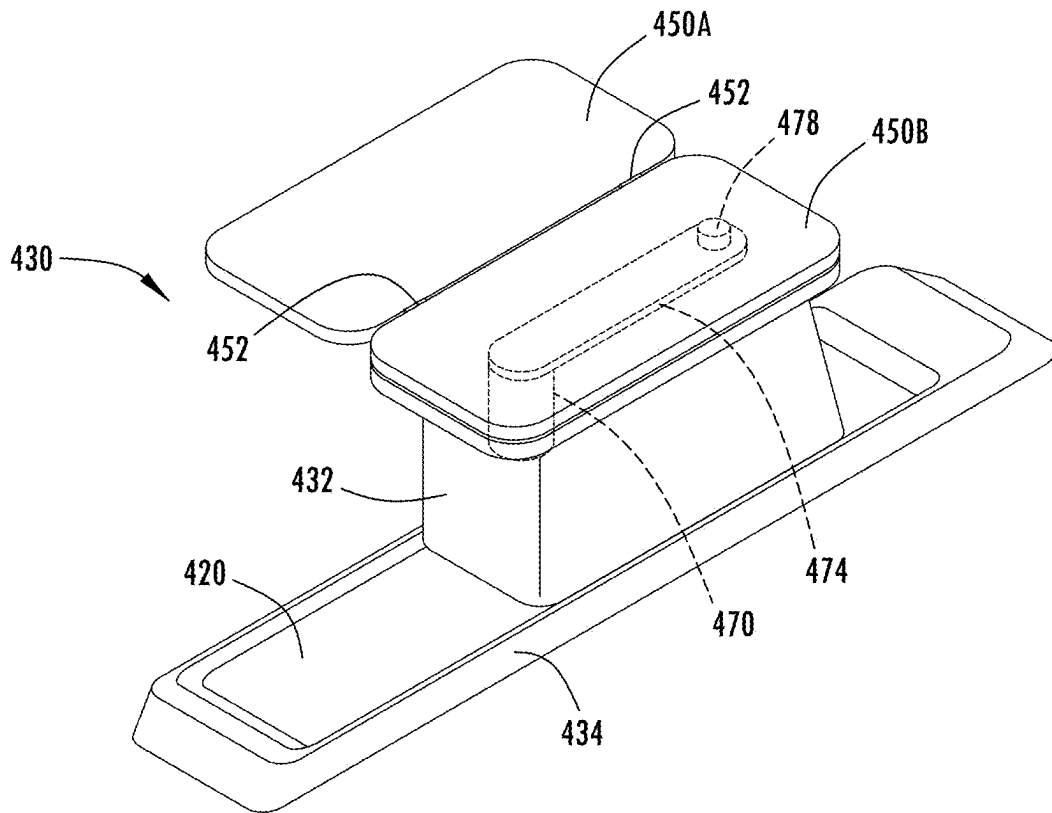


FIG. 19B

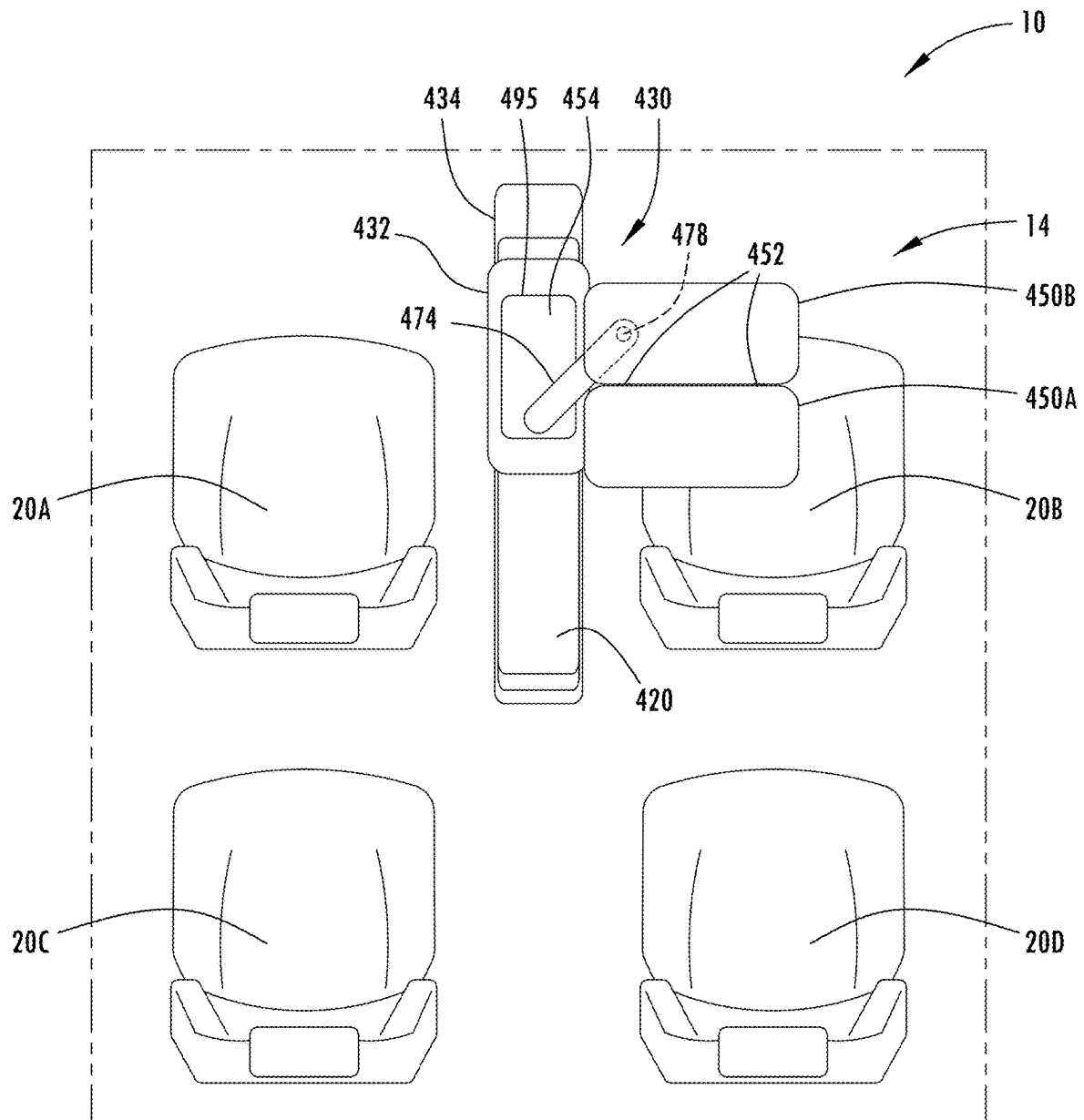


FIG. 20A

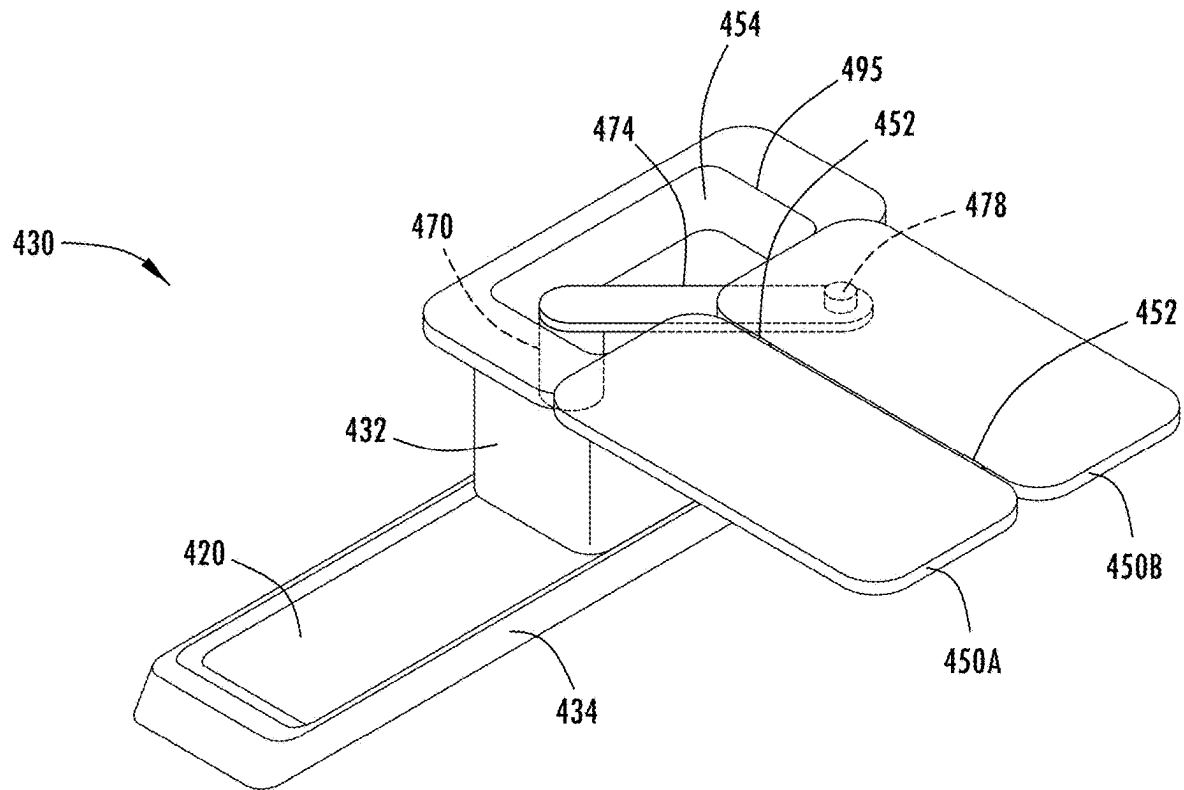


FIG. 20B

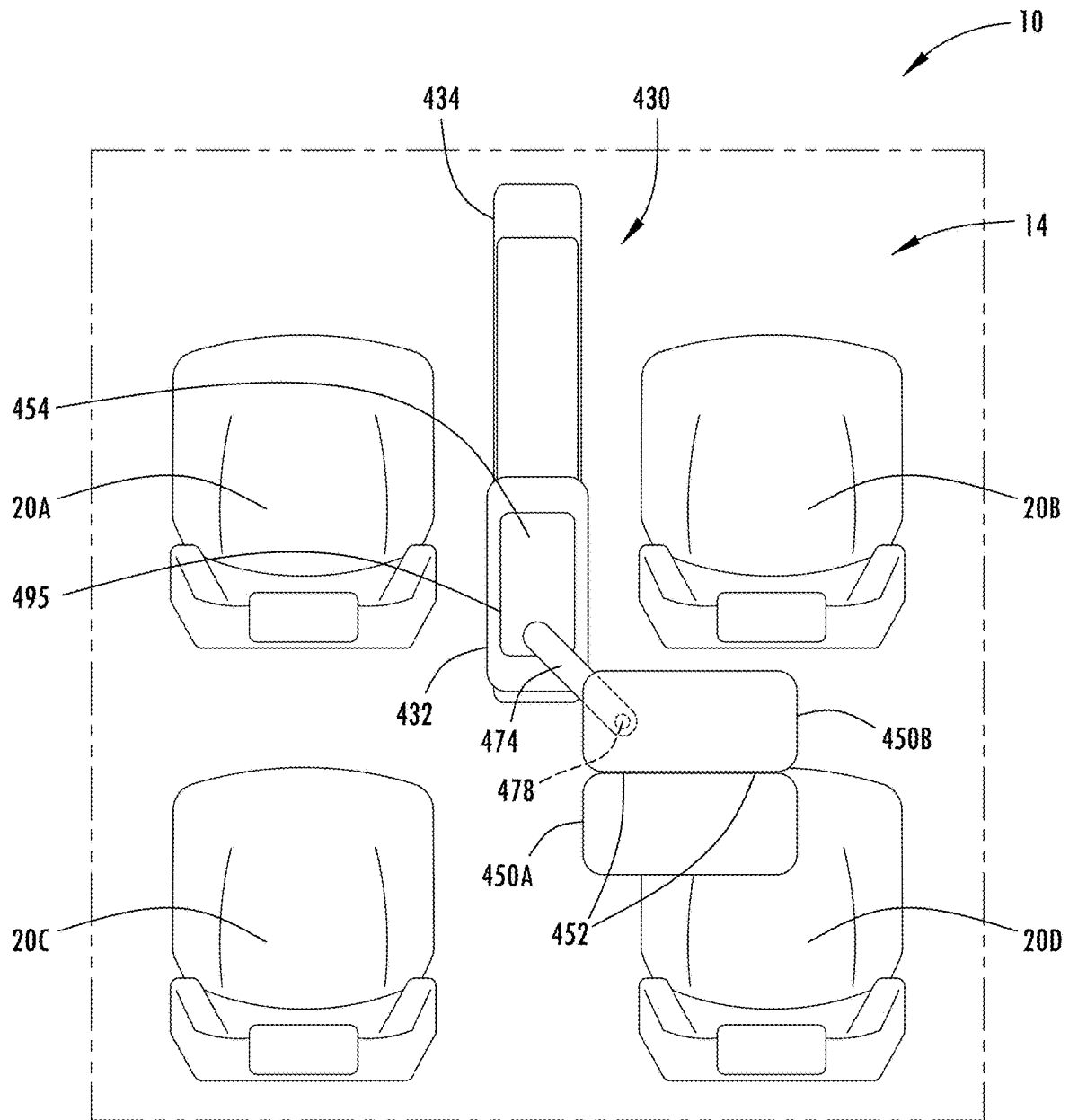


FIG. 21A

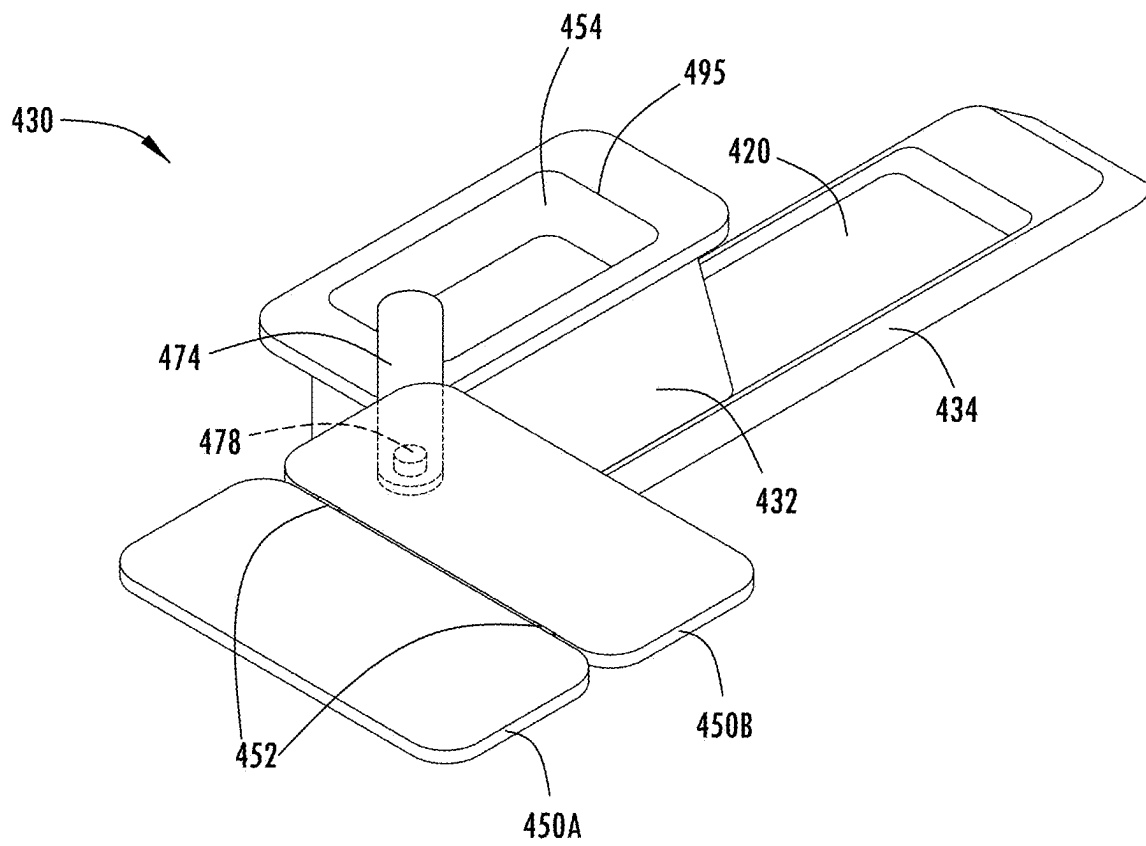


FIG. 21B

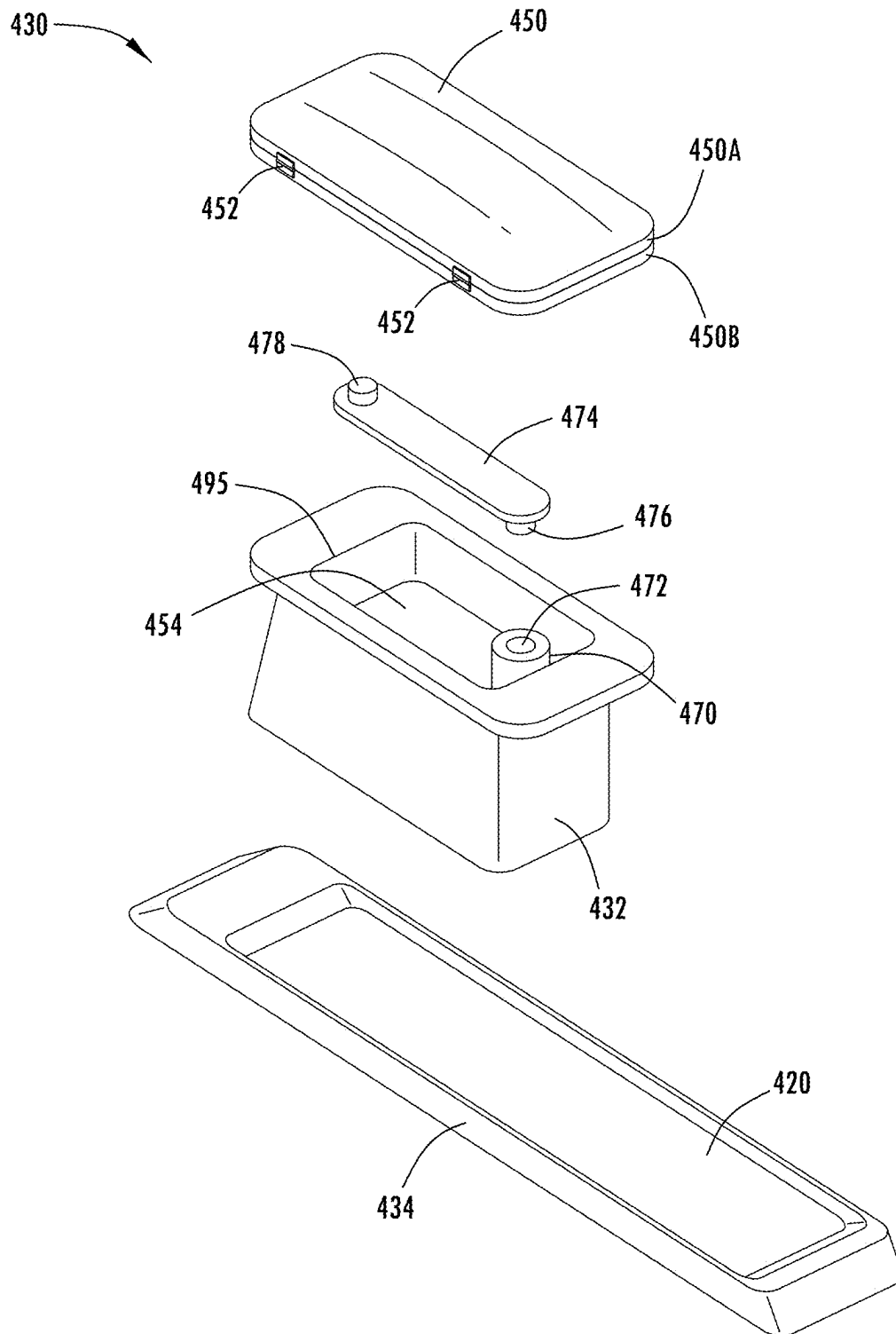


FIG. 22

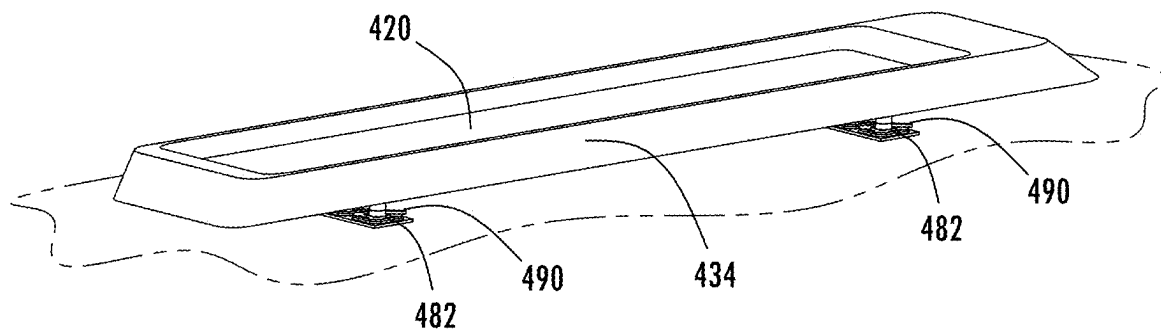


FIG. 23

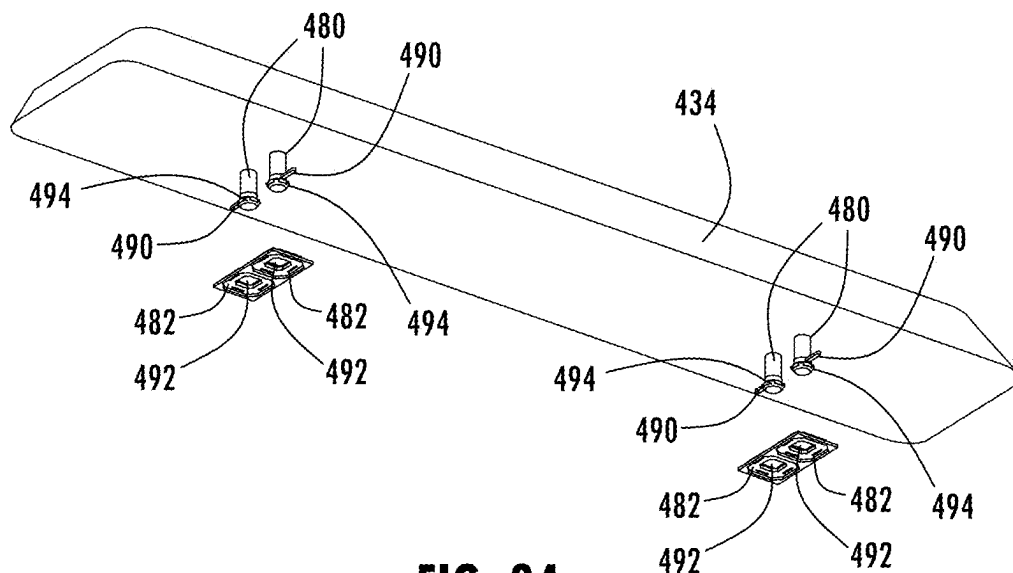


FIG. 24

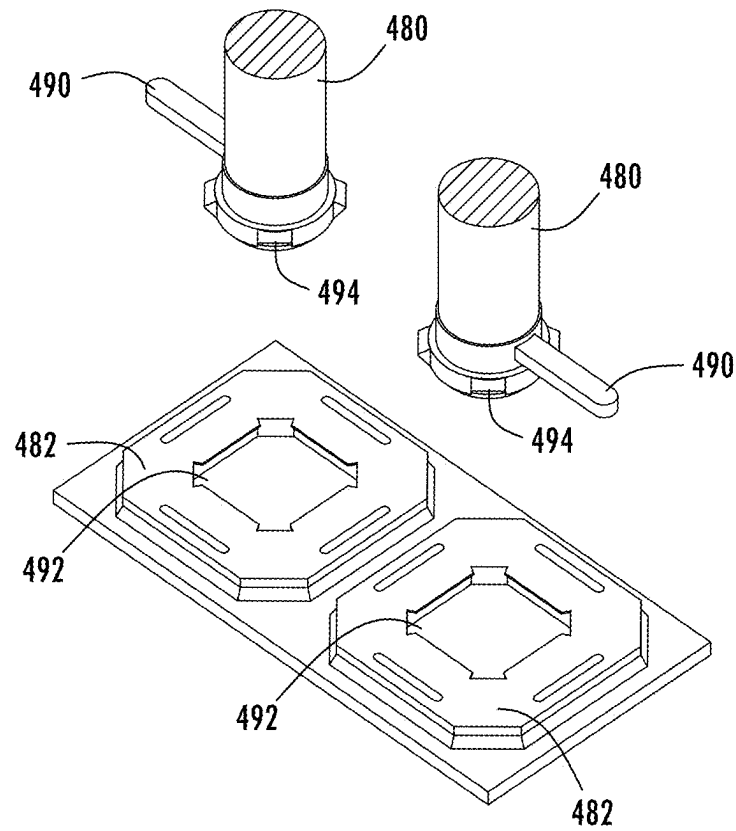


FIG. 25A

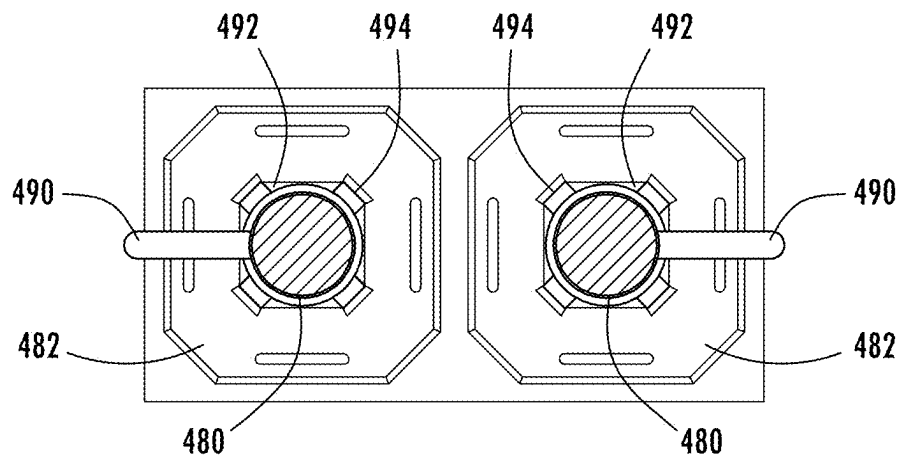


FIG. 25B

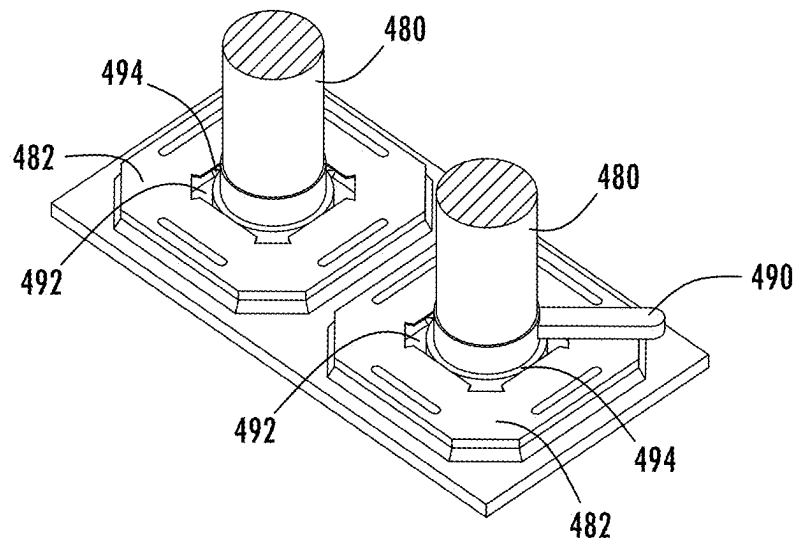


FIG. 26A

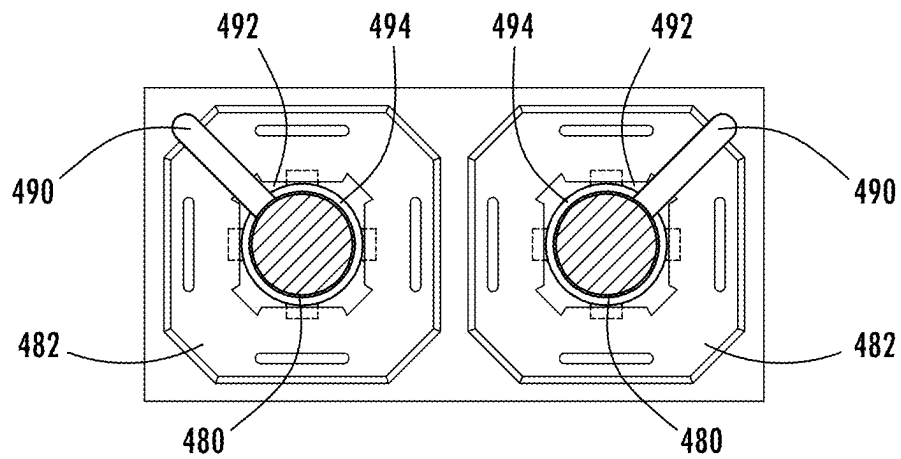


FIG. 26B

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VEHICLE CONSOLE HAVING COOLER ARRANGEMENT

FIELD OF THE DISCLOSURE

The present disclosure generally relates to consoles, and more specifically relates to a modular console for carrying items on a vehicle.

BACKGROUND OF THE DISCLOSURE

Motor vehicles are commonly equipped with storage compartments, such as consoles. For example, a center console is often installed between a driver seat and a passenger seat in a row of seating in a passenger motor vehicle. The center console typically has a main storage compartment with a lid which also commonly serves as an armrest and may include one or more drink holders and other storage compartments. It would be desirable to provide for a center console which offers enhanced features for use on a motor vehicle.

SUMMARY OF THE DISCLOSURE

According to a first aspect of the present disclosure, a vehicle console is provided. The vehicle console includes a base, a housing supported on the base and comprising a storage compartment, the storage compartment having side walls in an access opening, and a lid operatively coupled to the access opening of the housing and comprising a zipper for connecting the lid to the top end of the module, wherein the zipper may be unzipped to expose the access opening and may be zipped to close the lid on the access opening.

Embodiments of the first aspect of the present disclosure can include any one or a combination of the following features:

the lid comprises a cushion top that may serve as an armrest in a closed lid position;

the zipper comprises a first zipper portion that extends around a perimeter of the access opening of the housing and a second zipper portion that extends around a perimeter of the lid;

a cooler insert container configured to be disposed into the storage compartment through the access opening of the housing, the insert container configured for holding one or more items to be cooled;

the cooler insert container comprises a plurality of cavities for receiving one or more items to be cooled and a second plurality of cavities for receiving a cooling medium;

the console comprises a center console;

at least one cupholder formed on the module;

a tray configured to be supported on the base;

the lid is an insulated armrest;

an air flow duct extending through the housing and providing an outlet for passing airflow; and

the air duct extends through the base and into the module.

According to a second aspect of the present disclosure, a vehicle center console is provided. The vehicle center console includes a base, a housing supported on the base and comprising a storage compartment, the storage compartment having side walls and an access opening, a lid operatively coupled to the top end of the housing and comprising a zipper for connecting the lid to the top end of the module, wherein the zipper extends around a perimeter of the access opening of the module, wherein the zipper may be unzipped to expose the access opening and may be zipped to close the

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lid on the access opening, and a cooler insert container configured to be disposed into the storage compartment through the access opening of the housing, the cooler insert container configured for holding one or more items to be cooled.

Embodiments of the second aspect of the present disclosure can include any one or a combination of the following features:

the lid comprises a cushion top that may serve as an armrest in the closed position;

the lid is an insulated armrest;

the cooler insert container comprises a plurality of cavities for receiving one or more items to be cooled and a second plurality of cavities for receiving a cooling medium;

an air flow duct extending through the housing and providing an outlet for passing airflow;

the air duct extends through the base and into the module;

the zipper comprises a first zipper portion that extends around a perimeter of the access opening of the housing and a second zipper portion that extends around a perimeter of the lid;

at least one cupholder formed on the module; and

a tray configured to be supported on the base.

These and other features, advantages, and objects of the present disclosure will be further understood and appreciated by those skilled in the art by reference to the following specification, claims, and appended drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

In the drawings:

FIG. 1 is an upper side perspective view of a cabin interior of a motor vehicle equipped with a front center console and a rear center console, according to one example;

FIG. 2A is a rear perspective view of the cabin interior showing a first embodiment of a modular center console configured with module holders to hold removable modules;

FIG. 2B is a rear perspective view of the cabin interior showing the modular center console equipped with front and rear modules, according to one example;

FIG. 3 is an enlarged exploded rear perspective view of the module center console illustrated in FIG. 2B;

FIG. 4A is a lower rear perspective view of a module removed from the rear module holder of the modular center console shown in FIG. 3;

FIG. 4B is an enlarged lower rear perspective view of the module assembled onto the module holder as seen in FIG. 4A;

FIG. 5 is a cross-sectional view taken through line V-V of FIG. 4B showing the removable connection of the module onto the module holder, according to one example;

FIG. 6 is a schematic side view of the modular center console shown in FIG. 2B further illustrating the air ventilation flow path;

FIG. 7A is a perspective view of a removable ride share service module configured to be assembled onto the center console shown in FIG. 2A;

FIG. 7B is a perspective view of a removable soft cooler module configured to be assembled onto the center console shown in FIG. 2A;

FIG. 7C is a perspective view of a removable biometric security lock box module configured to be assembled onto the center console shown in FIG. 2A;

FIG. 7D is a perspective view of a removable open storage module configured to be assembled onto the center console of FIG. 2A;

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FIG. 7E is a perspective view of a removable wireless speaker module configured to be assembled onto the center console of FIG. 2A;

FIG. 7F is a perspective view of a removable trash bin module configured to be assembled onto the center console of FIG. 2A;

FIG. 8A is a rear perspective view of a second embodiment of a center console having a swappable base housing;

FIG. 8B is a rear perspective view of the center console shown in FIG. 8A with the armrest lid in a deployed table position, according to one example;

FIG. 8C is a rear perspective view of the center console shown in FIG. 8A with the armrest lid further positioned in an open lid position;

FIG. 9 is an enlarged exploded rear perspective view of the center console illustrated in FIG. 8A;

FIG. 10 is a schematic side view of the center console illustrated in FIG. 8A;

FIG. 11A is a rear perspective view of a third embodiment of a center console having road trip features and accessories;

FIG. 11B is a rear perspective view of the center console illustrated in FIG. 11A with the armrest lid in an open lid position;

FIG. 12 is an enlarged exploded rear perspective view of the center console shown in FIG. 11B with the armrest lid in the open console position;

FIG. 13 is a schematic side view of the center console shown in FIG. 11A;

FIG. 14A is a rear perspective view of a fourth embodiment of a center console equipped with media features and accessories;

FIG. 14B is a rear perspective view of the center console illustrated in FIG. 14A with the armrest lid in the open lid position;

FIG. 14C is a rear perspective view of the center console illustrated in FIG. 14B with an armrest lid in the open lid position and the media storage compartment empty;

FIG. 14D is a rear perspective view of the center console illustrated in FIG. 14A with a media accessory located in the media storage compartment;

FIG. 15 is an enlarged exploded rear perspective view of the center console illustrated in FIG. 14A;

FIG. 16 is a front perspective view of the center console illustrated in FIG. 14A further illustrating an audio speaker;

FIG. 17 is a schematic side view of the center console illustrated in FIG. 14A;

FIG. 18A is a top view of the cabin interior having a fifth embodiment of a center console with an armrest lid configured to provide a positional deployable table;

FIG. 18B is a rear perspective view of the center console illustrated in FIG. 18A with armrest lid in a closed lid position;

FIG. 19A is a top view of the cabin interior showing the center console of FIG. 18A with the armrest lid deployed in a first deployable table position;

FIG. 19B is a rear perspective view of the center console shown in FIG. 19A with the table in the first deployable table position;

FIG. 20A is a top view of the cabin interior of FIG. 18A showing the center console with the armrest lid deployed in a second deployed table position;

FIG. 20B is a rear perspective of the center console shown in FIG. 20A with the armrest lid in the second deployed table position;

FIG. 21A is a top view of the cabin interior showing the center console of FIG. 18A with the armrest lid further deployed in a third deployed table position;

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FIG. 21B is a rear perspective view of the center console shown in FIG. 21A with the armrest lid deployed in the third deployed table position;

FIG. 22 is an enlarged exploded perspective view of the center console illustrated in FIG. 18A;

FIG. 23 is an upper perspective view of the base of the center console shown in FIG. 18A with male leg connectors for connecting the base to the vehicle floor;

FIG. 24 is a lower perspective view shown in FIG. 23 showing the male leg connectors positioned to engage female floor connectors;

FIG. 25A is an enlarged perspective view of the male leg connectors aligned to engage the female connectors on the vehicle floor;

FIG. 25B is a top view of the male leg connectors engaged in the female floor connectors in the unlocked position;

FIG. 26A is an upper perspective view of the male leg connectors engaged in the female floor connectors in a locked position; and

FIG. 26B is a top view of the male leg connectors engaged in the female floor connectors shown in the locked position.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

Reference will now be made in detail to the present preferred embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numerals will be used throughout the drawings to refer to the same or like parts. In the drawings, the depicted structural elements are not to scale and certain components are enlarged relative to the other components for purposes of emphasis and understanding.

As required, detailed embodiments of the present disclosure are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the disclosure that may be embodied in various and alternative forms. The figures are not necessarily to a detailed design; some schematics may be exaggerated or minimized to show function overview. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the present disclosure.

For purposes of description herein, the terms “upper,” “lower,” “right,” “left,” “rear,” “front,” “vertical,” “horizontal,” and derivatives thereof shall relate to the concepts as oriented in FIG. 1. However, it is to be understood that the concepts may assume various alternative orientations, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

The present illustrated embodiments reside primarily in combinations of method steps and apparatus components related to a modular vehicle console. Accordingly, the apparatus components and method steps have been represented, where appropriate, by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the

art having the benefit of the description herein. Further, like numerals in the description and drawings represent like elements.

As used herein, the term “and/or,” when used in a list of two or more items, means that any one of the listed items can be employed by itself, or any combination of two or more of the listed items, can be employed. For example, if a composition is described as containing components A, B, and/or C, the composition can contain A alone; B alone; C alone; A and B in combination; A and C in combination; B and C in combination; or A, B, and C in combination.

In this document, relational terms, such as first and second, top and bottom, and the like, are used solely to distinguish one entity or action from another entity or action, without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.

As used herein, the term “about” means that amounts, sizes, formulations, parameters, and other quantities and characteristics are not and need not be exact, but may be approximate and/or larger or smaller, as desired, reflecting tolerances, conversion factors, rounding off, measurement error and the like, and other factors known to those of skill in the art. When the term “about” is used in describing a value or an end-point of a range, the disclosure should be understood to include the specific value or end-point referred to. Whether or not a numerical value or end-point of a range in the specification recites “about,” the numerical value or end-point of a range is intended to include two embodiments: one modified by “about,” and one not modified by “about.” It will be further understood that the end-points of each of the ranges are significant both in relation to the other end-point, and independently of the other end-point.

The terms “substantial,” “substantially,” and variations thereof as used herein are intended to note that a described feature is equal or approximately equal to a value or description. For example, a “substantially planar” surface is intended to denote a surface that is planar or approximately planar. Moreover, “substantially” is intended to denote that two values are equal or approximately equal. In some embodiments, “substantially” may denote values within about 10% of each other, such as within about 5% of each other, or within about 2% of each other.

As used herein the terms “the,” “a,” or “an,” mean “at least one,” and should not be limited to “only one” unless explicitly indicated to the contrary. Thus, for example, reference to “a component” includes embodiments having two or more such components unless the context clearly indicates otherwise.

Referring to FIG. 1, a wheeled automotive or motor vehicle 10 is generally illustrated configured with passenger seating for transporting one or more passengers including a driver of the vehicle 10 and for further transporting one or more items onboard the vehicle 10. The motor vehicle 10 has a vehicle body 12 that generally defines a cabin interior 14. The cabin interior 14 may contain various features and trim components. The cabin interior 14 is shown having an

arrangement of front row passenger seats including a first seat assembly 20A configured to seat a driver and a second seat assembly 20B configured to seat a passenger. The first and second seat assemblies 20A and 20B may be configured as captain seats, for example, laterally spaced apart and separated by space that includes a front center console 30 that may be configured according to various embodiments shown and described herein. The motor vehicle 10 illustrated in the example also includes additional seating such as a second row of seating including third and fourth seat assemblies 20C and 20D spaced apart and separated by a rear center console 30A that may be configured according to the various embodiments shown and described herein, and a third row of seating including fifth and sixth seat assemblies 20E and 20F. The motor vehicle 10 may include additional seating as is common for a large van or bus. It should be appreciated that the vehicle 10 may be a motor vehicle, such as a wheeled car, truck, SUV, van or bus, for example, or an airplane, train, boat or other vehicle capable of transporting one or more passengers and items.

The motor vehicle 10 is illustrated with a front row of seating having the first seat assembly 20A located on one lateral side (e.g., the left side) of the first row and vehicle rearward of a dashboard 26 and a steering wheel 22 and configured for seating a driver of the vehicle 10. The first seat assembly 20A includes a seat base and a seat back. The second seat assembly 16 is shown positioned vehicle rearward of the dashboard 26 on the opposite lateral side (e.g., right side) of the front row of seating and has a seat base and a seat back configured for seating a passenger. The vehicle body 12 has a plurality of side doors 16 that allow the driver and passengers to enter and exit the cabin interior 14 to access the seat assemblies 20A-20F. In the example shown, the driver seat assembly 20A is located adjacent to a lateral left side door 16 and the passenger seat assembly 20B is located adjacent to an opposite lateral right side door 16. Passengers may gain access to or exit from the second and third rows of seat assemblies 20C-20F via the rear side doors 16.

The cabin interior 14 is shown equipped with a front center console 30 located in the space between the driver first seat assembly 20A and the passenger second seat assembly 20B in the front first row of seating. In addition, the cabin interior 14 is also shown having a rear center console 20A located in a space between the third and fourth seat assemblies 20C and 20D in the middle second row of seating. It should be appreciated that the center console 30 or 30A may be located at various locations within the vehicle 10, such as between other adjacent seat assemblies. For example, a console may be located in a space between the fifth and sixth seat assemblies 20E and 20F in the rear third row of seating.

The vehicle console 20 is shown and described herein in accordance with various different embodiments for providing a console adjacent to one or more vehicle seating assemblies in the motor vehicle 10 that offers various features and accessories. For example, the vehicle console 30 may provide storage space such as one or more compartments for storing one or more items on the motor vehicle 10. In addition, the vehicle console 30 may provide an armrest for a driver or passenger of the motor vehicle 10 to rest their arm upon. Further, the vehicle console 30 may include a worktable that provides a worksurface upon which a user may utilize for purposes of work, for example. Further, the vehicle console 30 may include other features and accessories, such as various types of removable modules that may be added or removed from the console housing and

may include different bases and housings that may be added or replaced with other bases and console housing, according to the various embodiments shown and described herein. Additive Console

Referring to FIGS. 2A-7F, a vehicle console **30** is illustrated according to a first embodiment configured with first and second module holders **36A** and **36B** that are each configured to receive and hold different selectable removable modules. With particular reference to FIG. 2A, the vehicle console **30** is shown installed vehicle rearward of a rearward facing display device **24**, such as a touchscreen display, that is shown mounted on or proximate to the vehicle dashboard **26** at the front end of the cabin interior **14**. The vehicle console **30** has a housing **32** that serves as a spline and also may serve as a base in this embodiment that is shown fixedly attached onto a supporting structure such as the underlying vehicle floor **18**, according to one example. It should be appreciated that the housing **32** may include a separate base attached to the housing **32** and vehicle floor **18** and may otherwise be fixedly attached onto another supporting structure on the motor vehicle **10**. The housing **32** may be mounted to the vehicle floor **18** via fasteners such as screws or bolts or may be welded or may be otherwise connected to the vehicle floor **18** in a manner to be removable.

The housing **32** generally defines a first module holder **36A** generally shown on the front side facing a front side of the motor vehicle **10** and a second modular holder **36B** generally defined on the rear side and facing a rear side of the motor vehicle **10**. The first module holder **36A** is generally defined along a perimeter by a generally rectangular upstanding sidewall **40A** that extends upward from a holder floor **38A** and may include rounded corners. The first module holder **36A** is generally shaped to receive a base **60** of a removable module which may be selected from a plurality of available modules **70A-70F** as shown in FIGS. 7A-7F, for example. Formed on the outside of the sidewall **40A** is a retention detent **46A**. It should be appreciated that a retention detent **40A** may be provided on each of the opposite lateral left and side sides of the module holder **36A** and on the front side of the module holder **36A**. The retention detents **46A** are configured to be engaged by a securing clip **62** on a removable module to removably secure the module fixedly on the module holder **36A**. In addition, the module holder **36A** has a plurality of electrical contacts **42A** shown provided in the holder floor **38A** and exposed on top to allow for contact with contacts in a module. The electrical contacts **42A** may provide a supply of electrical power from the vehicle battery and may allow for data communication with vehicle electronics via electrical contacts **78** on the bottom side of the module installed in the module holder **36A**.

The second module holder **36B** is provided on the rear side of the housing **32** and likewise includes a generally rectangular upstanding sidewall **40B** extending around a perimeter from the holder floor **38B** and may include rounded corners shaped to conform to the shape of the base **60** of a removable module to be received in the second module holder **36B**. The second module holder **36B** likewise includes a retention detent **46** shown formed in an outer sidewall. It should be appreciated that the retention detent **46** may be provided in each of the opposite lateral left and right sidewalls of the module holder **36B** as well as on the rear wall of the module holder **36B** and is each configured to engage a securing clip on the removable module to fixedly hold the module in place on the second module holder **36B**. The second module holder **36B** likewise includes a plurality

of electrical contacts **42B** shown provided in the module floor **38B** which may be configured to provide electrical power from the vehicle battery and may allow for data communication with vehicle electronics via electrical contacts on the bottom side of the removable module.

The second module holder **36B** is shown having a ventilation or airflow duct port **44B** provided in the holder floor **38B** which may receive airflow for heating ventilation and air conditioning (HVAC) from the motor vehicle **10**. The ventilation duct port **44B** is configured to align and operably couple with a ventilation or airflow duct **76** provided in the removable module **70B** which in turn is operably coupled with an airflow outlet **74** at the rear end or the removable module **70B** as seen in FIGS. 2B and 6. As such, the removable module **70B** receives ventilation, heating and air conditioning, airflow from the HVAC system in the motor vehicle **10** and passes and outputs the airflow rearward of the module **70B**, such as for directing the airflow towards the rearward seated passengers seated in the cabin interior **14** in the middle or rear rows of seating.

Referring to FIGS. 2A, 2B and 3, the vehicle console **30** is generally shown having a first removable module **70A** installed and fixedly attached to the first module holder **36A** and a second removable module **70B** shown installed and fixedly attached to the second module holder **36B**. In this example, the first and second modules **70A** and **70B** may be selected from a group or a plurality of available modules **70A-70F** shown in FIGS. 7A-7F. Each removable module **70A-70F** is configured having a modular base **60** with securing clips and electrical contacts configured to engage the first and second module holders **36A** and **36B**. In the example shown, the first removable module **70A** is a soft cooler module that provides a soft upper housing with thermal insulated walls for forming a cooler, and the second module **70B** is a service module that includes a user input device in the form of a touchscreen display **72** that allows passengers as users to interact with services onboard the motor vehicle **10**, for example.

Referring to FIG. 5, the fixed connection of the removable module **70A** on the module holder **36A** is shown. The removable module **70A** has a securing clip **62** which includes a resilient finger having an inward extending tooth **63**. The tooth **63** is biased inward and configured to engage the retention detent **46A** which is shown configured as a slot formed in the outer wall of the module holder **36A**. As such, the tooth **63** engages the retention detent **46A** with a bias force to hold the removable module **70A** fixed in place. It should be appreciated that the retention detent **46B** likewise receives a tooth on a securing clip to retain a removable module on the module holder **36B**. It should be appreciated that a passenger may simply disconnect the securing clip **62** from engagement with the retention detents **46A** or **46B** by forcing the tooth **63** outward from the retention detent **46A** to easily remove the module and replace the module with another module.

The vehicle console **30** generally includes an upstanding tower **33** extending upward between the first and second module housings **36A** and **36B** which may define a central storage compartment **54** on the interior and a pair of cup holders **40A** near the front top end. The storage compartment **54** has a top access opening **95** that allows access to the compartment space. Extending over the access opening **95** of compartment **54** is a pivotable armrest lid **50** which in the rearward horizontal position covers the top access opening **95** of the storage compartment **54** and may serve as an armrest for the driver or passenger seated in the driver or passenger seat assemblies **20A** and **20B**, respectively. The

armrest lid **50** is shown as a pivoting lid having pivot pins **52** on opposite lateral sides that allow the armrest lid **50** to pivot between the horizontal rearward position covering the access opening **95** of the storage compartment **54** as seen in FIG. 3, and an upstanding or forward rotated position that exposes the access opening **95** of the storage compartment **54**.

The vehicle console **30** may further include a hanger **56** which is illustrated in FIG. 3, according to one example. The hanger **56** is generally formed on a rear upstanding wall of the tower **33** above the second module holder **36B** and provides a hook upon which one or more items may be hung, such as a strap on a purse or grocery bags, for example. The hanger **56** is particularly useful when there is no module installed in the second module holder **36B**. The hanger **56** is generally formed as an upstanding hook, according to the example shown. However, other hangers having other shapes and sizes may be employed. It should be appreciated that another hanger **56** may be provided on the front side above the first module holder **36A**, according to another example.

Referring to FIGS. 4A and 4B, the electrical contacts **79** on the bottom side of the base **60** of an exemplary removable module **70A** are shown configured to align with and connect with the electrical contacts **42A** or **42B** on the respective module holder **36A** or **36B** shown in FIG. 3 to close circuit electrical circuitry to receive a supply of electrical power and allow data communication. It should be appreciated that the power and data communication electrical contacts shown in FIG. 3 may communicate via electrical circuitry **78** with a power and data control module **64**. The electrical power may be supplied via power wires or wireless via induction, for example. The data communication may occur via wired or wireless communication such as Bluetooth®. This may enable the removable module retained in a module holder to power devices, such as a display, a cooling device, a heating device, media devices such as a gaming device, for example, or provide power and data communication outlets for connecting to other electronic devices, for example.

Referring to FIGS. 7A-7F, a plurality of selectable removable modules **70A-70F** are illustrated, according to one example. It should be appreciated that the vehicle console **30** is configured to receive and retain any of the selected removable modules **70A-70F** within either of the first and second module holders **36A** and **36B**. The group of selectable removable modules includes the first removable module **70A** shown in FIG. 7A configured as a soft cooler module, which may have a soft upper wall and top and may include a zipper top or other closure. The first removable module **70A** includes a rigid base **60** having a conforming shaped surface on the bottom that generally conforms to the shape of the module holder. The second selectable removable module **70B** shown in FIG. 7B is configured as a service module generally having a display such as a touchscreen display **72** that serves as a user input and output human machine interface (HMI) device, and an airflow duct **76** and airflow outlet **74**. A user may access or interface with the vehicle and associated electronics via the touchscreen display **72** to communication ride share information, vehicle service information, gaming information, and other forms of communication. The third removable module **70C** shown in FIG. 7C is configured as a biometric security lockbox module. Module **70C** is shown having a lid **80** on the top side with a handle **82** that may be grasped by the hand of a user. Included on the top side is a fingerprint sensor **84** which may optically scan the user's finger to look for a biometric fingerprint match and may have an actuator to open the

security lockbox when the biometric match occurs. The fourth removable module **70D** is shown configured as an open storage module in FIG. 7D. The open storage module **70D** generally includes a somewhat rectangular container having an open top end. The fifth removable module **70E** shown in FIG. 7E is configured as a wireless speaker module which may employ wireless communication to communicate with an electronic device and broadcast audio sound. The wireless communication may include Bluetooth® communication or other forms of wireless communication available onboard the motor vehicle. The sixth removable module **70F** shown in FIG. 7 is configured as a trash bin module which may include a lid on top covering a trash compartment in the closed lid position, according to one example.

It should be appreciated that any of the removable modules **70A-70F** may be selected and assembled onto and removably connected to either of the first and second module holders **36A** and **36B**. It should further be appreciated that the first and second module holders **36A** and **36B** may each include the airflow duct and the electrical contacts shown in module **70B**. The replaceable modules **70A-70F** may or may not include each of the airflow duct outlets and electrical contacts, depending on the features provided on each individual removable module. For example, the second removable module **70B** shown in FIG. 7B has an airflow duct **76** and airflow outlet **74**, whereas other of the removable modules do not include airflow ducts passing therethrough. Similarly, the second removable module **70B** has electronics related to the display and other possible electronics, whereas other of the removable modules may or may not require a power and data communication connection. It should further be appreciated that other types of removable modules having various other features may be employed in the vehicle console **30**, according to other examples.

Vehicle Service Console

Referring to FIGS. 8A-10, a vehicle console **130** is illustrated according to a second embodiment. The vehicle console **130** is shown as a center console positioned in the space between the driver seat assembly **20A** and passenger seat assembly **20B** in the front row of the motor vehicle **10**. In the example shown, the vehicle console **130** employs a swappable housing **132** that may connect to and disconnect from a base **134** and provides various service features that may be useful for vehicle services such as ride share services, delivery vehicle services, and providing or performing other services such as purchasing or obtaining items from a vending compartment showcase, acquiring cleaning items, such as for example disinfecting the motor vehicle **10** after use or before use, providing payment and advertising features, and other service related features.

The vehicle console **130** includes a housing **132** removably connected to an underlying base **134**. The base **134** is fixedly connected to the vehicle floor **18** or other support structure in the motor vehicle **10** with connectors which may include fasteners such as bolts or screws, or via welding and other connectors, for example. Accordingly, the housing **132** is equipped with various console features and may be detached and disassembled from the base **134** and may be swapped out with another console housing.

The vehicle console **130** has a plurality of compartments for providing storage space for storing one or more items. For example, a tray **120** is provided on the front end of the housing **132** which may be defined on a periphery by an upstanding wall. Disposed on top of the tray **120** is a removable tray **122** which is shown having a generally

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rectangular configuration, according to one example. The tray **120** and the removable tray **122** disposed on top thereof may hold one or more items.

The vehicle console **130** further includes a main storage compartment **154** with an access opening **195** on the top rear end. The access opening **195** is shown in FIG. **8A** covered by an armrest lid **150** which serves as a lid and an armrest and also serves as a deployable worktable. The armrest lid **150** includes a first table **150A** integrally connected via one or more hinges **152** to a second table **150B**. In FIG. **8A**, the armrest lid **150** is shown in a fully stowed folded position and covering the access opening **195** to the main compartment **154**. The first table **150A** is configured to pivot outward about the one or more hinges **152** on one lateral side of the access opening **155** as seen in FIG. **8B** to provide an expanded area worktable made up of both the first and second tables **150A** and **150B**. In addition, the second table **150B** may further be rotated and folded about the one or more hinges **152** outward to uncover the access opening **195** for the main compartment **154**. In this open lid position, the second table **150B** provides a reduced size worksurface.

The vehicle console **130** further includes another compartment **155** vehicle forward of the main compartment **154** that may act or may serve as a garbage container, according to one example. The garbage container configuration may include a base **180** disposed within the bottom surface of the compartment **155** and a rim or bezel **126** located on the top surface along a perimeter of the top access opening **195** which is configured to receive and hold a liner for a garbage bag in place. Pivotally connected to the bezel **126** is a lid **124** which is connected via one or more hinges **128** that may open and close the lid **124** to allow access or closure to the garbage container configured compartment **155**.

The housing **132** further includes a vending compartment **110** shown configured on one lateral side which provides a showcase for obtaining, such as purchasing, items held in the showcase. The vending compartment **110** includes an opening for accessing the vending compartment from the lateral side and has a lower wall **112** to hold items in place and an open upper window **113** to expose the items for presentation and access. Accordingly, a passenger in the motor vehicle **10** may access one or more items from the vending compartment **110**. It should be appreciated that while a vending compartment **110** is shown on one lateral side of the housing **132**, another vending compartment may likewise be located on the opposite lateral side of the housing **132** and other vending compartments may be located on the front and/or rear ends of the housing **132**, according to other examples.

The housing **132** further includes a pair of cupholders **148** shown provided on the top surface between the main compartment **154** and the garbage container compartment **155**. It should be appreciated that various other compartments may be provided on or in the housing **132** with various shapes and sizes, according to other examples.

Connected at the rear end of the center console **130** is a human machine interface (HMI) that may operate as a user input device and user output device shown in the form of a touchscreen display **172**. The touchscreen display **172** may be integrally formed or attached to or proximate the rear end of the housing **132** and generally faces vehicle rearward towards the rearward row of seating. As such, passengers seated in the middle second row of seating may view and access the touchscreen display **172** to input data and commands and to view or otherwise receive output data or information. For example, a passenger may interact with the touchscreen display **172** to obtain, such as purchases, one or

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more items from the showcase or compartments on the vehicle, such as to purchase an item on a ride share service vehicle, for example.

In addition, an airflow duct outlet **174** is provided at the rear end of the housing **132** to provide airflow such as HVAC airflow, rearward towards the middle second row of seating. It should be appreciated that the airflow duct outlet **174** is operably coupled to an airflow duct **176** that extends through the housing **132** as shown in FIG. **10** and operably couples to an airflow duct **178** that is incorporated into the heating, ventilation and air conditioning (HVAC) system of the motor vehicle **10**.

Referring to FIGS. **9** and **10**, the swappable connection of the housing **132** onto the base **134** is further illustrated, according to one example. The base **134** is shown having a pair of female connectors **182** formed on opposite front and rear ends. The bottom of the housing **132** has a pair of male connectors **184** configured with a shape and size to extend into and engage the female connectors **182** to lock the housing **132** onto the base **134**. The connection of the female and male connectors **182** and **184** may include a snap-lock connection or a twist-lock connection, for example, which is removable. As such, a user may remove a housing **132** from the base **134** and may swap the housing **132** out for a different housing simply by disconnecting the housing **132** from the base **134** and attaching the replacement housing to the base **134**. This removable connector arrangement provides for a swappable center console on the motor vehicle **10**.

Road Trip Console

Referring to FIGS. **11A-13**, a vehicle console **230** is illustrated on the motor vehicle **10** according to a third embodiment. The vehicle console **230** is shown configured as a center console located in the space between the driver seat assembly **20A** and the passenger seat assembly **20B** in the front row of seating of the motor vehicle **10**. The vehicle console **230** is configured with various features that enable portability with vehicle road trips and allow for a housing of the vehicle console **230** to be swapped out for a different or replacement console housing, according to one example. The vehicle console **230** includes a base **234** fixedly connected to the vehicle floor **18** or other supporting structure. The connection of the base **234** to the vehicle floor **18** may be achieved with the use of fasteners, such as screws or bolts, or may include welding, for example. As such, the base **234** is intended to be fixed in place on the vehicle floor **18** while the remainder of the vehicle console **230** may be swapped out in this example.

Removably connected to the top side of the base **234** is a housing **232**. The housing **232** is configured to be removable and may be swapped out with a replacement housing on top of the base **234**. The housing **232** may include various compartments for holding and storing items. For example, the vehicle console **230** has a main storage compartment **254** with an access opening **295** on the top side thereof. The main compartment **254** is shown having a generally rectangular or cube-shaped configuration. Disposed on top of the housing **232** and covering the access opening **295** of the main compartment **254** is a flexible armrest lid **250** which may cover the access opening **295** of the main compartment **254** as seen in FIG. **11A** or may be opened to expose and allow access to the main compartment **254** in FIG. **11B**. The main compartment **254** may include an insulated compartment having thermal insulated walls, such as walls reinforced with foam insulation or other thermal insulation. The armrest lid **250** likewise may include thermal insulation **251**, such as foam insulation or other insulation. As such, the

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main compartment **254** is configured in this example to operate as a thermally insulated container such as a cooler to keep items stored therein thermally isolated with low heat exchange from the surrounding environment.

The armrest lid **250** is configured with a zipper that is made up of a first zipper piece **252A** operatively coupled to a second zipper piece **252B**. The first zipper piece **252A** extends around a perimeter of the access opening **295** of the main compartment **254** on top of the housing **232**. The second zipper piece **252B** extends around the perimeter of the armrest lid **250**. The first zipper piece **252A** is matingly engaged to the second zipper piece in the closed lid position and may be unzipped to open up the compartment **254** by separating the first zipper piece **252A** from the second zipper piece **252B** with a sliding motion. As such, a plurality of teeth of the first and second zipper pieces **252A** and **252B** are engaged and locked together when the armrest lid **250** is in the closed lid position.

Disposed within the main compartment **254** is a container having walls defining a compartment shown as a cooler insert container **210**. The cooler insert container **210** may be portable and insertable into and removable from the main compartment **254** and may be fixed in place with releasable fasteners. The cooler insert container **210** is shown having a plurality of ice packs **212** which may be arranged and spaced to provide openings for receiving items, such as drink containers **214**. As such, the ice packs **212** operate to thermally cool the drink containers **214**, in the example shown. As such, the cooler insert container **212** may be disposed within the main compartment **254** and the first and second zipper pieces **252A** or **252B** may be zipped closed to close the armrest lid **250** to provide for an insulated cooler surrounding the cooler insert container **210**. As such, passengers in the motor vehicle **10** may access thermally cooled beverages or other cooled items.

In addition, the housing **232** includes a pair of drink holders **248** on a top surface at a forward end thereof and an airflow duct outlet **274** at the rear end thereof. The airflow duct outlet **274** may be operatively coupled to an airflow duct **276** extending through the housing **232** as seen in FIG. **13**. The airflow duct **276** may, in turn, be operatively coupled to a vehicle airflow duct **278** that transfers air from the vehicle heating, ventilation and air conditioning (HVAC) system. As such, HVAC airflow is able to flow out the airflow outlet **274** at the rear end of the vehicle console **230** and into the cabin interior **14**.

The vehicle console **230** further includes a tray **220** generally defined by an upstanding peripheral wall extending forward at the front end of the housing **232**. Disposed on the tray **220** is a removable tray **222** which may include a pair of handles and may rest on top or be fixedly attached to tray **220**. As such, additional items may be held on the tray **220** and the removable tray **222**.

The base **234** has a pair of female connectors **282** at opposite front and rear ends thereof. The bottom of housing **232** has a pair of male connectors **284** which extend into the female connectors **282** as seen in FIG. **13**. The male connectors **284** may engage the female connectors **282** in a snap-fit configuration or a twist-lock configuration to lock the housing **232** onto the base **234**. As such, a passenger or user may unsnap or twist the housing **232** to remove it from the base **234** and may swap the housing **232** out for a different housing and connect it to the base **234**. Accordingly, the vehicle console **230** advantageously provides for a vehicle console having a cooler for vehicle road trips. The housing **232** of the center console **230** may be swappable by replacing the housing **232** with a replacement housing.

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Media Console

Referring to FIGS. **14A-17**, a vehicle console **330** is illustrated according to a fourth embodiment. The vehicle console **330** is shown in this embodiment as a vehicle center console arranged in the space between the driver seat **20A** and the passenger seat **20B** in the front row of seating of the motor vehicle **10**. The vehicle console **330** is generally located vehicle rearward of the front display device **24** which is shown assembled on the dashboard **26**, in this example. The vehicle console **330** includes a base **332** fixedly connected to a supporting structure on the motor vehicle **10**, such as the vehicle floor **18**. The base **332** may be connected to the vehicle floor **18** via fasteners, such as bolts or screws, or may be welded to the vehicle floor **18**, for example. The vehicle console **330** further includes a housing **334** removably connected to the top surface of the base **332**. In one example, the housing **334** may be swappable with a replacement housing by disconnecting a snap-fit connection or a twist-lock connection and reinserting a new console housing onto the base **332**.

The housing **334** generally defines a plurality of compartments that may be used as storage compartments to store one or more items onboard the motor vehicle **10**. The housing **332** includes a main compartment **354** having a top access opening **395** to allow access to the main compartment **354**. An armrest lid **350** is shown pivotally connected to the top side of the housing **332** to close the access opening **395** in a closed lid position as seen in FIG. **14A** and to pivot to an open forward position to expose the access opening **395** of the main compartment **354** as seen in FIG. **14B**. It should be appreciated that the armrest lid **350** may be hingedly connected via one or more hinges **352** along the top forward edge of the access opening **395** on the housing **334** adjacent to the access opening **395** of the main compartment **354**. The armrest lid **350** may also serve also as an armrest having a cushioned top surface in the closed lid position shown in FIG. **14A**. Further, the armrest lid **350** may serve as a table in the forward rotated open position as seen in FIG. **14B**.

Disposed within the main compartment **354** are a plurality of power and data communication outlet ports. As seen in FIG. **14C**, included are a plurality of electrical outlets **370** are provided on the inner wall of the main compartment **354** for supplying electrical power to power one or more electrically powered devices. The electrical outlets **370** may provide direct current (DC) voltage electrical power supplied by the vehicle battery, or the DC power may be converted to provide an alternating current (AC) voltage. It should be appreciated that the outlets **370** may be configured with standard power connectors for receiving a plug with terminals to receive the electrical power. In addition, one or more data ports **372** are also provided on the inner wall of the main compartment **354**. The data ports **372** may include one or more USB connector ports, for example, or other data communication ports such as phone or computer connection ports. As such, one or more electrically powered and/or data communication devices may be stowed within the main compartment **354** as shown in FIGS. **14B** and **14D** and may be plugged into one or more of the power or data ports **370** or **372** to be powered and/or to be charged and/or communicate data.

As seen in FIG. **14B**, a couple of electronic media devices in the form of game controllers **360** are shown stowed in the main compartment **354**, according to one example. The game controllers **360** may be electrically powered and/or charged by plugging electrical cables into the power ports **370**. In addition, the game controllers **360** may be connected to electrical cables to provide data communication with the

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communication ports 372 to play video games or communicate other data. As seen in FIG. 14D, a game control module 362 is shown disposed within the main compartment 354. The game control module 362 may likewise be connected to one or more of the power ports 370 to power or charge the game control module 362 and may provide data communication with the motor vehicle 10 via one or more of the data communication ports 372.

Referring to FIG. 15, it should be appreciated that the main compartment 354 may include an inductive charging mat 382 installed on top of a floor console mat 380 inside of the main compartment 354. The inductive charging mat 382 may provide electrical power wirelessly via induction to charge one or more electrically powered devices or batteries in the electrically powered devices onboard the motor vehicle while held within the main compartment 354. The inductive charging may use charging coils in the compartment that inductively couples with charging coils in the device or batteries. A pair of brackets 384 are shown for holding one or more electronic devices, such as for positioning the devices to be charged in position with the inductive charger to maintain sufficient charging coupling.

The vehicle console 330 further includes a media audio device, such as an audio speaker 390, shown located at the front end of the housing 332. The audio speaker 39 may be a wired speaker wired to one or more vehicle electronic, according to one example. It should be appreciated that the audio speaker 390 may be a wireless speaker according to another example that may communicate with one or more electronic devices via wireless communication such as Bluetooth® or other wireless communication.

In addition, the motor vehicle 10 is shown equipped with a hanger 356 for hanging one or more electrical devices, such as a pair of earphones or headphones 358 as seen in FIG. 14C. As such, the headphones 356 may be used with the game controller 360 and the control module 362 to provide audio sound and to provide a microphone, if needed. The vehicle console 330 is positioned vehicle rearward of a front display device 24 which is shown located on the dashboard 26 of the vehicle 10. The front display device 24 may be a touchscreen display and may provide user inputs and may provide outputs, such as video outputs for outputting media, such as gaming data or gaming videos. As such, one or more passengers seated in the motor vehicle 10 may utilize the vehicle console 330 as a media console to provide media accessories for storage and use on the motor vehicle 10.

The base 334 of vehicle console 330 is illustrated in FIG. 15 having a pair of female connectors in the form of receptacles 382 located at or proximate to opposite front and rear ends of the base 334. The female receptacles 382 are configured to be engaged with a pair of male connectors 384 located on the bottom surface of the housing 332 as seen in FIGS. 15 and 17. As such, the male connectors 384 engage the female connectors 382 to fixedly hold the housing 332 in place on the base 334. It should be appreciated that the male connectors 384 may include snap connectors that snap into place or twist connectors that twist-lock onto and off the base 334. As such, a passenger or user may disconnect the housing 332 from the base 334 and may swap the housing 332 out for a replacement housing and reconnect the replacement housing on the base 334.

Accordingly, the vehicle console 330 advantageously provides for a media console having various media-related features and accessories for use on the motor vehicle 10. The

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vehicle console 330 may be swapped out and replaced with a different replacement console, such as another media console.

Positionable Worktable

Referring to FIGS. 18A-26B, a fifth embodiment of a vehicle console 430 in the cabin interior 14 of the motor vehicle 10 is illustrated. The vehicle console 430 is shown and described herein as a center console supported on the vehicle floor 18 and located in the space between the driver seat 20A and the passenger seat 20B in the front row of the motor vehicle 10. The center console 430 may be connected to the vehicle floor 18 or other supporting structure via fasteners, such as bolts or screws, or otherwise welded or connected to the supporting structure. In this embodiment, the center console 430 includes a deployable table that may be deployed to a plurality of positions within the cabin interior 14 of the vehicle 10. In addition, the vehicle console 430 has a housing 432 that may be removably connected to a base 434 fixed on the vehicle floor 18 and may be swapped out for a different replacement housing.

The vehicle console 430 includes the base 434 having removable fasteners for removably connecting the base 434 to the vehicle floor 18 or a support on the vehicle floor 18. The base 434 generally has a planar upper surface 420 upon which the housing 432 is located. The housing 432 is positioned on the planar surface 420 along a longitudinal track and may move such as by sliding forward and rearward on top of surface 420. The housing 432 generally defines a storage compartment 454 and has an access opening 495 on the top thereof. The storage compartment 454 has a space that may hold one or more items. Disposed across the top of the access opening 495 of the storage compartment 454 as an armrest lid 450 which serves as a lid and an armrest also serves as a deployable table. The armrest lid 450 is shown in the closed position in FIGS. 18A and 18B fully covering the access opening 495 of the storage compartment 454. The armrest lid 450 has a first table 450A pivotally connected to a second table 450B via one or more hinges 452. As such, the first table 450A may pivot about the one or more hinge 452 along a lateral side from the second table 450B to provide an expanded worksurface table as seen in FIGS. 19A and 19B.

The first worktable 450A is connected to the storage compartment 454 via an upstanding support post 470 and a horizontal support arm 474 which pivots relative to the support post 470. The support arm 474 has a pivot pin connector 478 extending vertically upward which engages a recess in the lower surface of the second worktable 450B. The opposite end of the control arm 474 has a lower extending pivot pin 476 that engages a slot 472 at the top surface of the support post 470. As such, the support arm 474 is able to pivot relative to the support rod 470 to rotate the armrest lid 450 as a worktable amongst a plurality of positions. For example, the worktable 450A in a deployed position as seen in FIG. 19A can be rotated to a second position as seen in FIGS. 20A and 20B in which the table moves from a position proximate to the driver seat assembly 20A to a position proximate and forward of the front passenger seat assembly 20B. This enables a passenger seated in the front passenger seat to use the worktable. The worktable 450A may further be rotated to a third position as seen in FIG. 21A. In this position, the housing 432 slides vehicle rearward along the track on planar upper surface 420 towards the rear second row of seating. In a third position, the deployed worktable 450A is presented proximate to and forward of the rear seat assembly 20D. This enables a passenger seated in the rear seat assembly 20D to have

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access to the worktable. It should further be appreciated that the worktable **450A** may further be rotated and positioned proximate to the opposite lateral side seat assembly **20C** in the second row of seating. This enables a passenger seated in the rear seat assembly **20C** to have access to the worktable. As such, the vehicle console **430** advantageously provides a multipositional workstation that may be moveably positioned and presented for use to a plurality of seated passengers in the vehicle **10**.

Referring to FIGS. **23A-26B**, the connection of the base **434** to the vehicle floor **18** is illustrated, according to one example. The bottom side of the base **434** has a plurality of downward extending legs **480** that form a connector assembly. In the example shown, four legs **480** are provided, with two legs located towards a front end of the base **434** and the other two legs located towards a rear end of a base **434**. Each of the legs **480** has a shaped foot **494** on the bottom end thereof and a locking lever **490** operatively coupled to a cam surface associated therewith. In this example, the vehicle floor **18** has a plurality of matching receptacles **482** each having a shaped opening **492** for receiving the shaped feet **494** on legs **480**. According to one example, the shaped surface has an octagonal shape and the locking feet **494** are also octagonal in shape. The shaped feet **494** may be inserted into the openings **492** and the lever **490** may be rotated so that the shapes of the feet and receptacle misalign and lock the feet **494** within the opening **492**. It should be appreciated that the locking lever **490** may engage a cam surface to maintain a frictional engagement upon turning a quarter of a turn (90°), for example. However, it should be appreciated that the lever **490** may be turned more than a quarter turn or less than a quarter turn, according to other examples.

The connecting legs **480** and locking feet **494** may further include an electrically conductive material such that they may serve as electrical connectors for supplying power and/or data communication through the base **434** to the housing **432**. As such, the connecting feet may also serve as power and communication connecting features.

The vehicle console **430** may be easily assembled onto the vehicle floor **18** or other vehicle structure using the connector arrangement described herein. For example, a vehicle console **430** may be removed easily from the vehicle floor **18** and replaced with another swappable vehicle console simply by disconnecting the four legs and feet to remove the console and reconnecting the four legs and feet on the replacement console.

It should be appreciated that the various vehicle consoles shown and described herein may be standalone units in some embodiments, and may include interchangeable modules, according to further embodiments, and may further be swappable to replace the console from the motor vehicle as shown and described herein. Features of any of the center consoles may be employed on other of the center consoles shown and described herein. While modern OEMs of passenger vehicles would currently warn against using the deployable table or positionable table while the vehicle is in motion due to safety concerns, it is anticipated that technology and the regulatory framework may evolve in the future to where such an activity is safe and permissible.

It is to be understood that variations and modifications can be made on the aforementioned structure without departing from the concepts of the present disclosure, and further it is to be understood that such concepts are intended to be covered by the following claims unless these claims by their language expressly state otherwise.

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What is claimed is:

1. A motor vehicle comprising:

a supporting structure; and

a vehicle console comprising:

a base fixedly connected to the supporting structure, the base having a top supporting surface and one or more base connecting features and having a first airflow duct configured to transfer air from an HVAC system in the vehicle;

a housing having a bottom surface and one or more housing connector features configured to matingly connect with the one or more base connector features to connect the housing on the base and allow for the housing to be removed from the base, the housing comprising a storage compartment, the storage compartment having side walls and an access opening, wherein the housing has a second airflow duct operatively coupled to the first airflow duct at one end and an airflow duct outlet at an opposite end; and

a lid operatively coupled to the access opening of the housing and comprising a zipper for connecting the lid to the top end of the housing, wherein the zipper may be unzipped to expose the access opening and may be zipped to close the lid on the access opening.

2. The motor vehicle of claim 1, wherein the lid comprises a cushion top that may serve as an armrest in a closed lid position, and wherein the housing is configured as a replaceable module.

3. The motor vehicle of claim 2, wherein the zipper comprises a first zipper portion that extends around a perimeter of the access opening of the housing and a second zipper portion that extends around a perimeter of the lid.

4. The motor vehicle of claim 1 further comprising a cooler insert container configured to be disposed into the storage compartment through the access opening of the housing, the insert container configured for holding one or more items to be cooled.

5. The motor vehicle of claim 4, wherein the cooler insert container comprises a plurality of cavities for receiving one or more items to be cooled and a second plurality of cavities for receiving a cooling medium.

6. The motor vehicle of claim 1, wherein the console comprises a center console.

7. The motor vehicle of claim 1 further comprising at least one cupholder formed on the housing.

8. The motor vehicle of claim 1 further comprising a tray configured to be supported on the base.

9. The motor vehicle of claim 1, wherein the lid is an insulated armrest.

10. A motor vehicle comprising:

a supporting structure; and

a vehicle center console comprising:

a base fixedly connected to the supporting structure, the base having a top supporting surface and one or more base connecting features and having a first airflow duct configured to transfer air from an HVAC system in the vehicle;

a housing having a bottom surface and one or more housing connector features configured to matingly connect with the one or more base connector features to connect the housing on the base and allow for the housing to be removed from the base, the housing comprising a storage compartment, the storage compartment having thermal insulated side walls and an access opening, wherein the housing has a second

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airflow duct operatively coupled to the first airflow duct at one end and an airflow duct outlet at an opposite end;

a thermal insulated lid operatively coupled to the top end of the housing and comprising a zipper for connecting the lid to the top end of the housing, wherein the zipper extends around a perimeter of the access opening of the housing, wherein the zipper may be unzipped to expose the access opening and may be zipped to close the lid on the access opening; and

a cooler insert container configured to be disposed into the storage compartment through the access opening of the housing, the cooler insert container configured for holding one or more items to be cooled.

11. The motor vehicle of claim 10, wherein the lid comprises a cushion top that may serve as an armrest in the

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closed position, and wherein the housing is configured as a replaceable module.

12. The motor vehicle of claim 11, wherein the lid is an insulated armrest.

13. The motor vehicle of claim 10, wherein the cooler insert container comprises a plurality of cavities for receiving one or more items to be cooled and a second plurality of cavities for receiving a cooling medium.

14. The motor vehicle of claim 10, wherein the zipper comprises a first zipper portion that extends around a perimeter of the access opening of the housing and a second zipper portion that extends around a perimeter of the lid.

15. The motor vehicle of claim 10 further comprising at least one cupholder formed on the housing.

16. The motor vehicle of claim 10 further comprising a tray configured to be supported on the base.

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